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Analysis of neuroscience-inspired intervention restoring the plasticity of bimodal artificial neural networks in the context of critical learning periods

Master's thesis
in the field of COGNITIVE SCIENCE

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Abstract

This study explores a method for restoring plasticity in bimodal deep neural networks, drawing inspiration from recent advancements in neuroscience. The method presented here builds upon research into amblyopia, focusing on its onset and treatment methods, to highlight the parallels between the two research areas. Moreover, tools derived from recent studies on the analysis of critical learning periods in unimodal and bimodal deep neural networks assist in analyzing the dynamics of changes in the branches of the bimodal network that occur during the critical period and after applied intervention. Firstly, the study demonstrates that introducing corrupted (blurred) data to one branch of the bimodal network reduces the network's reliance on that branch's weights for predictions. This effect persists throughout the training process, even after removing the corrupted data. The study then correlates the degree of imbalance between the network's branches with the classification accuracy on the test dataset. This is observed both before and after the intervention, suggesting that models with balanced reliance on both branches achieve better classification accuracy. A key finding of this study is that the correction method improves classification accuracy on Fashion MNIST and CIFAR10 dataset following a deprivation period. This finding challenges existing claims that the generalization capability of bimodal deep models in the final training phase is determined early in the training process. Further experiments reveal that the choice of hyperparameters, such as a higher learning rate or increased weight decay, significantly influences the degree of imbalance in the network's reliance on its branches.

Keywords

plasticity, critical learning periods, multimodal deep artificial neural network, amblyopia, cataract, classification task, fisher information matrix, relative source variance

The title of the thesis in Polish

Analiza interwencji inspirowanej neuronauką przywracającej plastyczność bimodalnych sztucznych sieci neuronowych w kontekście krytycznych okresów uczenia się

Abstrakt

Niniejsza praca bada sposób przywracania plastyczności w bimodalnych głębokich sieciach neuronowych inspirowany ostatnimi wynikami w neuronauce. Zaprezentowana metoda śledzi badania nad amblyopią, jej powstawaniem i sposobami jej wyleczenia, aby wykazać zaobserwowane podobieństwa między dwoma obszarami badań, natomiast narzędzia zaczerpnięte z ostatnich prac nad analizą okresów krytycznych uczenia się w unimodalnych oraz bimodalnych głębokich sieciach neuronowych, służą pomocą przy analizie dynamiki zmian w gałęziach sieci bimodalnej, które zachodzą podczas okresu krytycznego i po zastosowanej interwencji. W pierwszej kolejności zaprezentowane zostało iż wprowadzenie korupcji do danych będących wejściem do jednej z gałęzi sieci bimodalnej, prowadzi do mniejszego polegania, podczas predykcji, na wagach tej gałęzi, w stosunku do drugiej gałęzi, w dalszym ciągu treningu, nawet wtedy gdy korupcja ta została usunięta. Następnie, stopień zaobserwowanej dysproporcji między gałęziami skorelowano ze skutecznością klasyfikacji na zbiorze testowym w finalnym etapie treningu przed i po interwencji, wskazując na lepszą skuteczność klasyfikacji gdy model polegał na obu gałęziach w równym stopniu. Głównym wynikiem tej pracy jest potwierdzenie że zaprezentowana metoda korekcji poprawia skuteczność klasyfikacji, na zbiorach Fashion MNIST oraz CIFAR10, po okresie deprivacyjnym, mimo istnienia prac twierdzących iż zdolność generalizacji bimodalnych głębokich modeli w finalnej fazie treningu jest przesądzona już na wczesnym etapie. Dodatkowe eksperymenty wskazują na stopień dysproporcji polegania na gałęziach sieci bimodalnej, w zależności od doboru hiperparametrów - większy learning rate lub większy weight decay prowadzą do większej dysproporcji.

Słowa kluczowe

plastyczność, krytyczny okres uczenia się, multimodalne głębokie sieci neuronowe, amblyopia, zaćma, zadanie klasyfikacji, macierz informacji fishera, relatywna wariancja źródeł

The title of the thesis in English

Analysis of neuroscience-inspired intervention restoring the plasticity of bimodal artificial neural networks in the context of critical learning periods

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Introduction

By drawing inspiration from neuroscience, the study of artificial neural networks has had the opportunity to be enriched by biological systems. Algorithm architectures inspired by neuroanatomy [McCulloch & Pitts 1943], methods of analyzing and processing neuronal signals [Rosenblatt 1958], and modeling cognitive processes [Fukushima 1980, LeCun et al. 1998] are just some of the significant examples of the intersection of these two fields. Despite loose inspiration, the artificial neuronal information processing systems proposed in machine learning seem to follow some principles of biological neuronal information processing systems. One of these principles is the existence of a critical learning period [Wiesel 1982] during which a temporary stimulus deficit, at early stage of learning, can impair the development of a skill. Research on human and animal amblyopia [Fong et al. 2021], a visual disorder also known as "lazy eye," has indicated that traditional treatments, which take into account the mechanisms leading to this disorder that occurred during a critical learning period phase, may be insufficient for full and lasting vision correction.

A similar phenomenon was observed in machine learning [Achille et al. 2019], where an unimodal DNN's ability to correctly classify images from a dataset decreased as the phase in which it was pretrained on deficit-induced images (blurred) was prolonged. The follow up research [Kleinman et al. 2022], revealed the existence of critical learning period phenomenon, crucial for accurately integrating signals from two modalities in bimodal DNN. Within the context of these cited articles, selected deficits originating from neuroscience, such as cataracts or strabismus - which have been identified as contributing factors to developmental amblyopia within the visual system — have been implemented into the DNN's learning process of an image classification task, to simulate deficit-like occurrence.

Considering the prior span of successful implementation of inspirations from neuroscience, the purpose of this study is to explore the effect of an intervention inspired by the amblyopia treatment method [Fong et al. 2021] on correcting plasticity in bimodal DNNs after a period of pretraining the model on deficit-induced data. The experiments and methodology detailed in the following chapters are an extension of the experiments and methodology, after proper translation, described in the mentioned articles about the existence of critical learning periods in bimodal DNNs [Kleinman et al. 2022], and in the article describing the intervention on the occurrence of amblyopia in animals [Fong et al. 2021]. Drawing from the latter, subsequent analogies between artificial and biological neural networks will be indicated, using tools proposed in the aforementioned studies on critical learning periods in DNNs.

To the author's knowledge, this study is the first to demonstrate the existence of a method to correct capabilities after the critical learning period of bimodal DNNs. Considering that current DNNs have limited plasticity when it comes to multiple training sessions on successive tasks, understanding and trying to counter the processes occurring during the critical period

may be crucial for future applications. In the subsequent chapters, the theoretical background of amblyopia and the critical learning period in the context of biological and artificial neural networks will be presented, followed by a detailed description of the experimental methods applied and the results obtained.

Chapter 1

Definitions

Here are the definitions of concepts gathered in one place according to which they will be understood in subsequent chapters.

Definition 1.0.1 (Cataract)

A cataract is the clouding (or spotting) of the lens in the eye, which makes it difficult for light rays to reach the retina, consequently leading to impaired sharpness, and this, in turn, can lead to amblyopia if not treated promptly enough. The primary treatment for cataract is the surgical removal of the clouded lens, which is then usually replaced with an artificial intraocular lens.

Definition 1.0.2 (Strabismus)

Strabismus is a condition in which the eyes do not align properly with each other when looking at an object. It can be caused by nerve injury or dysfunction of the muscles controlling the eyes, or by uncorrected refractive error. If untreated during childhood, it can lead to amblyopia in the misaligned eye. Treatment options include eyeglasses, prisms, vision therapy, or surgery.

Definition 1.0.3 (Amblyopia)

Often referred to as "lazy eye," amblyopia is a vision development disorder where at least one eye fails to achieve typical visual acuity, even with prescription eyeglasses or contact lenses. It usually begins during infancy or early childhood, indicating the existence of a critical period in the early stage of development. Typically caused by a lack of coordinated use of the two eyes, which might arise from one eye having a much different prescription than the other (cataract) or from one eye turning in or out (strabismus).

Definition 1.0.4 (Plasticity of a nervous system)

Plasticity is the ability of an nervous system to change its connections and behavior in response to new information, sensory experiences, development, damage, or dysfunction. This adaptability can manifest as the nervous system's capacity to reorganize by forming new neural connections, strengthening or weakening existing pathways, or reassigning functions to different neural circuits. Plasticity is foundational for learning, memory, recovery from brain injuries, and adaptation to changing environments or experiences. The brain's plastic nature is most pronounced during developmental stages but continues throughout life, albeit at a decreased rate in adulthood.

Definition 1.0.5 (Critical learning period in a nervous system)

A period in the nervous systems's development when it is particularly susceptible to learning certain skills or behaviors, and when it is highly plastic and capable of reorganizing in response to experience. After this period, it plasticity may be limited, which can hinder the acquisition

of new skills or adaptation to new situations. Critical learning periods have been identified in various areas of learning and development, such as language acquisition, sensory processing, and motor skills. The concept is often used in neuroscience and developmental psychology to explain why certain capabilities are more easily acquired during specific phases of life.

Definition 1.0.6 (Plasticity of an artificial neural network)

The ability of an artificial neural network to adapt to new information and utilize it for subsequent tasks. In the case of classification tasks, this would be the network's ability to effectively classify after prior learning, on subsequent classification tasks — as if each subsequent task were an initial task.

Definition 1.0.7 (Supervised learning)

Supervised learning refers to a paradigm in machine learning where a model is trained using a dataset consisting of input-output pairs. The primary objective is to train a function that, when given new and previously unseen input data, can predict their corresponding labels with high accuracy.

Let $\mathcal{X}$ denote the input space and $\mathcal{Y}$ the label space. Given a training set

$$\mathcal{D}_{\text{train}} = \{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$$

where $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$ for all $i = 1, \dots, N$, where $N \in \mathbb{N} \setminus \{0\}$.

The primary objective of supervised learning is to find a function $f : \mathcal{X} \rightarrow h[\mathcal{Y}]$ such that the difference between the predictions $f(x)$ and the actual labels y is minimized across all pairs (x, y) in $\mathcal{D}_{\text{train}}$. Moreover, this function should demonstrate predictive performance on the test set $\mathcal{D}_{\text{test}}$, which is utilized to assess the generalization ability of the model trained on $\mathcal{D}_{\text{train}}$.

In the above fragment, the function $h : \mathcal{Y} \rightarrow \mathbb{R}^{|\mathcal{Y}|}$ is a one-to-one function that transforms label data into values suitable for optimization steps. Whereas, $h[\mathcal{Y}] = \{h(y) \mid y \in \mathcal{Y}\}$.

During the training phase, the primary aim is to minimize the average value of a loss function L over the elements of $\mathcal{D}_{\text{train}}$. This function quantifies the discrepancy between the model's predictions and the actual labels:

$$\arg \min_f \frac{1}{N} \sum_{i=1}^N L(f(x_i), h(y_i))$$

Here, the loss function $L : h[\mathcal{Y}] \times h[\mathcal{Y}] \rightarrow \mathbb{R}_{\geq 0}$ evaluates the disparity (error) between the predicted label and its true counterpart.

Definition 1.0.8 (Classification Task with Deep Neural Network)

Classification is a particular type of supervised learning task where the label space is defined as

$$\mathcal{Y} = \{c_1, c_2, \dots, c_k\}$$

for some $k \in \mathbb{N} \setminus \{0, 1\}$, and h is a function that assigns to each label a different unit vector from $\mathbb{R}^{|\mathcal{Y}|}$.

In the context of classification, an Artificial Neural Network is a parametric model, represented as $f = f_\theta$ with parameters θ , that predicts the output $\hat{y} = f_\theta(x)$ based on the input x . This model consists of several layers of units, commonly known as neurons. Each i -th layer of these neurons processes its current input using the parameters θ_i , which are adjusted during training to minimize the discrepancy between the network's predictions and the actual labels. This adjustment is achieved through operations that determine the relationship between these parameters and the input.

Mathematically, the output of a layer in the network can be expressed as:

$$\mathbf{v}_{i+1} = a_i(\psi_i((\mathbf{W}_i, \mathbf{b}_i), \mathbf{v}_i))$$

where i indicates the layer index, $\mathbf{v}_i$ and $\mathbf{v}_{i+1}$ are the input and output of the layer, respectively, a_i is the activation function of the specified layer, $\theta_i = (\mathbf{W}_i, \mathbf{b}_i)$ are the parameters of the respective layer known as weights and bias, and ψ_i is a function that outlines the relationship between the parameters and the input of the current layer (e.g. $\psi_i((\mathbf{W}_i, \mathbf{b}_i), \mathbf{v}_i) = \mathbf{W}_i \cdot \mathbf{v}_i + \mathbf{b}_i$).

For classification tasks, the output from the ANN in our case employs a softmax function, converting its outputs into probability distributions over the set C . The predicted class label $\hat{c}$ is then given by:

$$\hat{c} = \underset{c \in \mathcal{Y}}{\operatorname{argmin}} \operatorname{softmax}(\hat{\mathbf{y}})$$

The softmax function compresses a k -dimensional vector of arbitrary real values into a k -dimensional vector of real values between 0 and 1 that sum up to 1. The label corresponding to the highest value in this vector is chosen as the predicted label $\hat{c}$.

We say that an artificial neural network is deep when it consists of a sufficiently large number of hidden layers - this name is conventional and does not require further specification.

Definition 1.0.9 (Fisher Information Matrix Trace)

The Fisher Information Matrix for a parametric model quantifies the information a given set of observable variables convey about the model's parameters.

Formally, for a parameter vector θ and variable Z , the Fisher Information Matrix $\mathbf{F}(\theta)$ (or simply $\mathbf{F}$) is defined as:

$$\mathbf{F}(\theta) = \mathbb{E}_Z \left[\left(\frac{\partial \log p_\theta(Z)}{\partial \theta} \right) \left(\frac{\partial \log p_\theta(Z)}{\partial \theta} \right)^T \right] \quad (1.1)$$

where $p_\theta(Z) = p(Z|\theta)$ denotes the likelihood of observing the variable Z given the parameters θ .

When computing the FIM for the probabilistic model $p_\theta(y|x)$, considering the input data distribution Q_X , the conditional target distribution $Q_y|x$, and assuming that the dependence of the joint distribution on the parameter θ arises exclusively from the class-conditional distribution while the marginal distribution remains θ -invariant, that is, $p_\theta(x, y) = p_\theta(y|x)p(x)$, we obtain:

$$\mathbb{E}_{x \sim Q_X} \mathbb{E}_{\hat{y} \sim p_\theta(y|x)} \left[\frac{\partial \log p_\theta(\hat{y}|x)}{\partial \theta} \frac{\partial \log p_\theta(\hat{y}|x)}{\partial \theta}^T \right] \quad (1.2)$$

while setting $Z = (X, Y)$, because:

$$\frac{\partial \log p_\theta(Z)}{\partial \theta} = \frac{\partial \log p_\theta(Y|X)}{\partial \theta} + \frac{\partial \log p(X)}{\partial \theta} = \frac{\partial \log p_\theta(Y|X)}{\partial \theta} \quad (1.3)$$

The assumption of parameter independence from the input distribution is justified since parameters are expected to track the conditional target distribution, aiming to learn the relationships between inputs and outputs rather than the inputs themselves.

Considering the aforementioned, the trace of the Fisher Information Matrix is expressed as:

$$\text{Tr}(\mathbf{F}) = \mathbb{E}_{x \sim Q_X, \hat{y} \sim p_\theta(y|x)} \left[\left\| \frac{\partial \log p_\theta(\hat{y}|x)}{\partial \theta} \right\|_2^2 \right] \quad (1.4)$$

Definition 1.0.10 (Source Variance)

Let j be the layer number in the main branch of a bimodal DNN f , and i be the unit number from that layer.

Given an input (a, b) consisting of inputs to two branches of the bimodal network, let's fix the input b , and using input a define the source A of elements close in distribution to element a such as $a' \sim p(a)$. For the sake of completeness, let's also define the source B which consistently returns the element b .

Given a source A from which data can flow, we define the Source Variance for this source, conditioned on element b for the unit i of the layer j as:

$$SV_{j,i}(A, b) = \text{Var}(f(A, B)_{j,i} \mid B = b) \quad (1.5)$$

which measures the variability of the response of the selected unit i of layer j based on the variability of the source, conditioned on an example b .

In other words, this measure quantifies the variability of a specific neuron in a specific layer when the input to one branch is fixed, and the input to the other is perturbed.

Definition 1.0.11 (Relative Source Variance (RSV))

Let j be the layer number in the main branch of a bimodal DNN f , and i be the unit number from that layer.

Given an input (a, b) consisting of inputs to the two branches of the bimodal network, use these inputs to define the sources A and B as sets of elements close in distribution to the elements a and b respectively.

Given two sources A and B , we define the Relative Source Variance for these sources with respect to the selected examples as:

$$RSV_{j,i}(a, b) = \frac{SV_{j,i}(A, b) - SV_{j,i}(B, a)}{SV_{j,i}(A, b) + SV_{j,i}(B, a)} \quad (1.6)$$

which measures the relative variability of the response of the selected unit i of layer j based on the variability of both sources.

In other words, this measure quantifies how much a specific neuron in a specific layer has a greater variance for one source compared to the other. The sum in the denominator serves solely to normalize the result to the range $[-1; 1]$.

Chapter 2

Related works

This chapter provides a summary of studies that form the foundation for the research presented in the following chapters. Each study will be introduced and analyzed, in the context of the concepts and assumptions that are fundamental to the experiments planned in this research. Every plot in this chapter was taken directly from the discussed article.

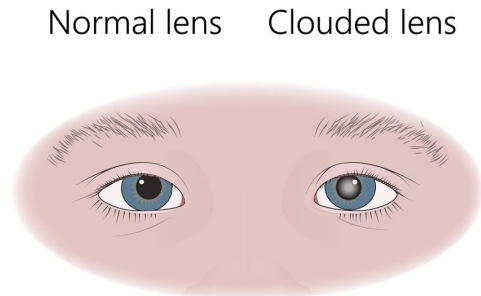
2.1. Postnatal development of the visual cortex and the influence of environment (Wiesel 1982)

Observations made in the field of neurobiology and psychology indicate the remarkable importance of early experiences for the development of visual functions. Children with congenital cataracts, despite their removal and proper correction, often experience significant and lasting visual deficit known as amblyopia. Similarly, behavioral studies have shown that animals raised in darkness or in an environment devoid of contours have similar visual function impairments. The presentation of amblyopia and its cause in the form of a cataract depicts Figure 2.1

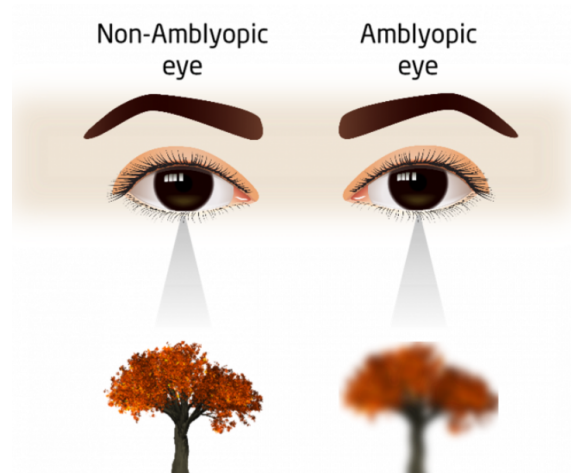
Pioneering research by Torsten N. Wiesel and David H. Hubel, spanning the 1950s to the 1970s and culminating in a Nobel Prize in the 1980s, shed light on the biological mechanisms behind these visual deficits. They highlighted that there is a period in development when visual experiences are crucial for the proper functioning of the visual cortex. Focusing on the visual cortex area of cats (and later confirming and expanding these results with deprivation studies on macaque monkeys), they discovered the specificity of the neurons' responses in the visual cortex to various spatial orientations of visual stimuli. This forms the fundamental mechanism by which these animals can detect and recognize object contours in their surroundings. The aforementioned observations regarding cataracts and being raised in darkness influenced the design of their experiments. In the early experiments, they decided to replicate the development process of the visual system during the deficit, by suturing the eyelids of young kittens. In this way, the nerve cells of the kittens' visual cortex received a light signal, to which changes in the level of scattering they are insensitive, in contrast to signals regarding contours, which this signal lacked.

The researchers noted that kittens with one eye sutured during their first three months exhibited blindness in that eye upon reopening, mirroring the effects seen in cataract development. However, there were no obvious changes in the ocular media, the retina, or the lateral geniculate nucleus that could explain this deficit. This deficiency was pinpointed primarily in the visual cortex, possibly where geniculate afferents meet cortical cells. These observations sug-

Cataract in infants



(a) A Comparative Illustration Showing a Normal Lens Versus a Clouded Lens.



(b) The left side shows a clear and sharp view from a Non-Amblyopic Eye, represented by a vivid tree. The right side shows a blurred, indistinct view from an Amblyopic Eye, highlighting the effects of amblyopia.

Figure 2.1: A Side-by-Side Depiction of Normal and Impaired Vision Due to Cataract and Amblyopia.

gest that restoring the functionality of the sensory organ is insufficient to regain full visibility in the eye that experienced the deficit, as changes occurred at the neural system level.

Furthermore, in the striate cortex of macaques, most cortical cells seemed to only respond when the non-sutured eye was stimulated. In a typical visual system, there isn't a significant dominance of one eye over the other in terms of cell activation in the striate cortex. In the context of research on the plasticity of the visual system, this is significant, as any disturbances in this balance may indicate disorders or changes in the way the brain processes visual information from each eye. Notably, the reorganization of connections and functions in response to new stimuli wasn't seen in mature animals but was present only during the initial months after birth, which presents closer Figure 2.2.

The experiments of the mentioned researchers highlighted the importance of brain's plasticity at an early stage of the nervous system's development and the existence of a "critical period", setting a new direction for research, emphasizing the role of early experiences in shaping visual functions. In the context of this master's thesis, it is essential to explore how the plasticity observed in biological neural networks can be a reference for artificial networks. The key questions are: Do artificial networks experience their own 'critical periods'? And if so, can we manipulate their plasticity to optimize performance at various learning stages?

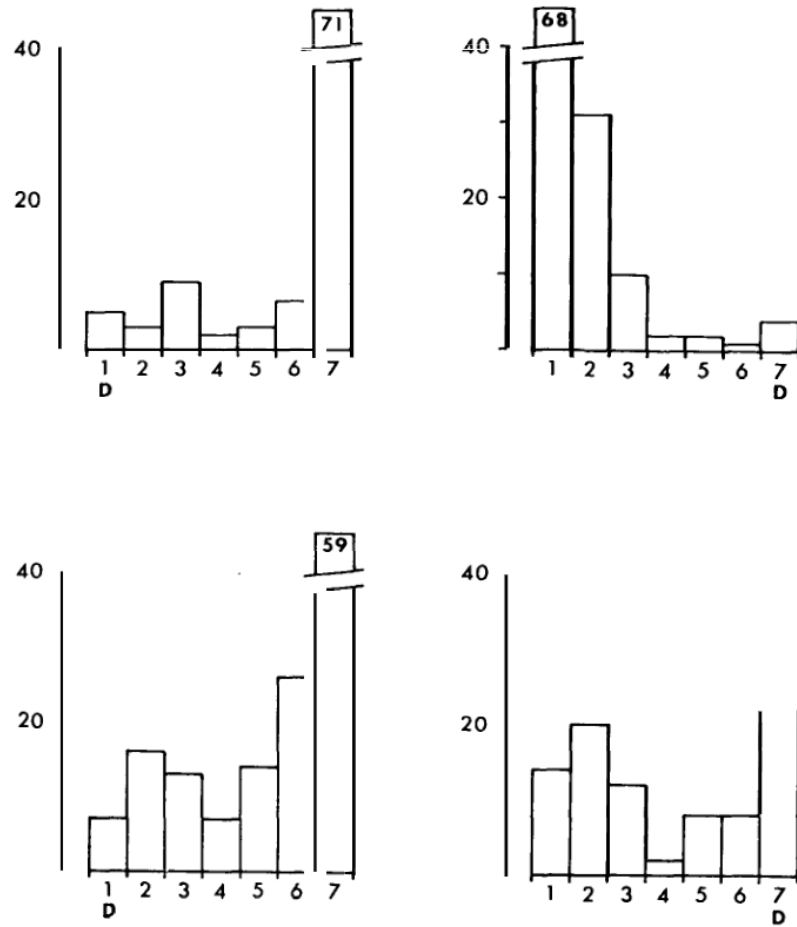


Figure 2.2: Figure adapted from Wiesel 1982, presents histograms counting neurons that have been activated in specific regions of the V1 visual cortex, layer IV (excluding IVC), related to the degree of activation for each eye:

1: Activation exclusively for stimuli presented to the contralateral eye.

7: Activation exclusively for stimuli presented to the ipsilateral eye.

2-6: Gradual sharing of activation for stimuli presented to both eyes.

The letter **D** indicates the side of the histogram that represents the deprived eye.

Upper left: Occlusion of the right eye at 2 weeks after birth for a duration of 18 months.

Upper right: Occlusion of the right eye at 10 weeks of age for a duration of 4 months.

Lower left: Occlusion of the right eye at the age of 1 year for a duration of 1 year.

Lower right: Occlusion of the right eye at age 6 for a period of 1.5 years.

The study demonstrates that limiting vision leads to an increased reliance on one eye the earlier it is applied during the development of the visual system.

2.2. Critical Learning Periods in Deep Neural Networks (Achille et al. 2019)

As previously introduced, in the context of biological neural networks, "critical learning periods" refer to specific phase in an organism's development during which a temporary sensory deficit can disrupt the development of certain skills. In this study, this concept was transferred to the domain of deep neural networks (DNNs).

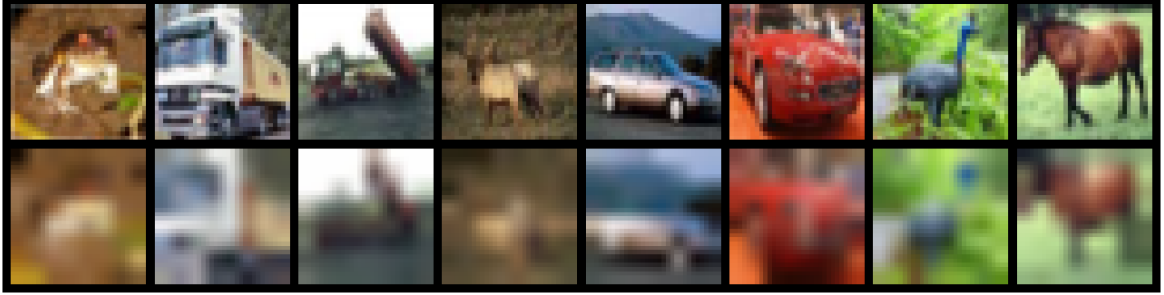


Figure 2.3: Presentation of the difference between data from the first and second phase based on the CIFAR10 dataset. The top row displays selected images from the CIFAR10 dataset, and the bottom row their blurred versions, intended to simulate the post-cataract effect.

Achille et al. 2019 investigates whether there are similar critical learning periods in training DNN models during a classification task. They explored if there are specific moments during the learning process when the models are more susceptible to adaptation to new data or tasks. To do this, they conducted a series of experiments, modifying the training data at different moments in the learning process. The main study involved dividing neural network training into two distinct phases. In the first phase, lasting a variable number of epochs, the DNN was trained on deficit-induced (blurred) images from a selected dataset; this phase was intended to represent the visual system’s adaptation to an environment in the case of a cataract occurrence. In the second phase, the DNN was trained for 160 epochs to allow it to converge, but this time on unaltered data from the same dataset; this phase was meant to represent the visual system’s adaptation after the lens’ clarity correction. The researchers observed that the longer the first phase lasted, the worse the classification performance was on the test set at the end of the second phase. Different architectures and optimization methods exhibited similar behavior concerning critical periods, suggesting that this is a universal phenomenon for DNNs and not specific to a particular architecture or optimization method. The results of the mentioned experiments, in conjunction with measurements of visual impairment in kittens, are illustrated in the accompanying Figure 2.4, while an example of the difference in data across different phases in Figure 2.3.

These findings indicate that the initial learning stage is crucial for the outcome of the training process, and that DNNs might lose their plasticity even due to a minor change in the training data distribution. This poses a significant challenge for transfer learning [Zhuang et al. 2020] or continual learning [Lange et al. 2021], where data on which a given model will be pretrained might prevent it from achieving optimal performance on a subsequent dataset.

To examine the dynamics of weight formation in DNNs, researchers analyze the Trace of the Fisher Information Matrix (TFIM). The researchers derived an interpretation of the FIM as the strength of synaptic connection in neural networks; that is, for a specific task, it indirectly measures the contribution of each individual weight, in the case of its perturbation, to the network’s output. Weights with low Fisher information have little impact on the network’s outcome and can be modified or "pruned" with minimal effect on its performance relative to the task. Consequently, Fisher information is proposed as a tool to assess the 'effective connectivity' within DNNs, which can be examined throughout the training process as shown in Figure 2.5a, depending on the duration of the first phase. The full Fisher Information Matrix

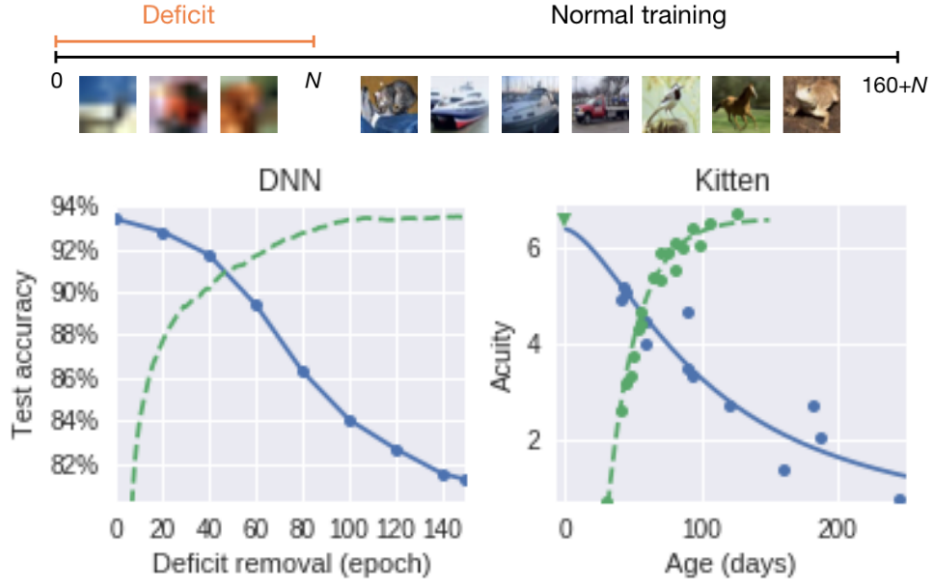
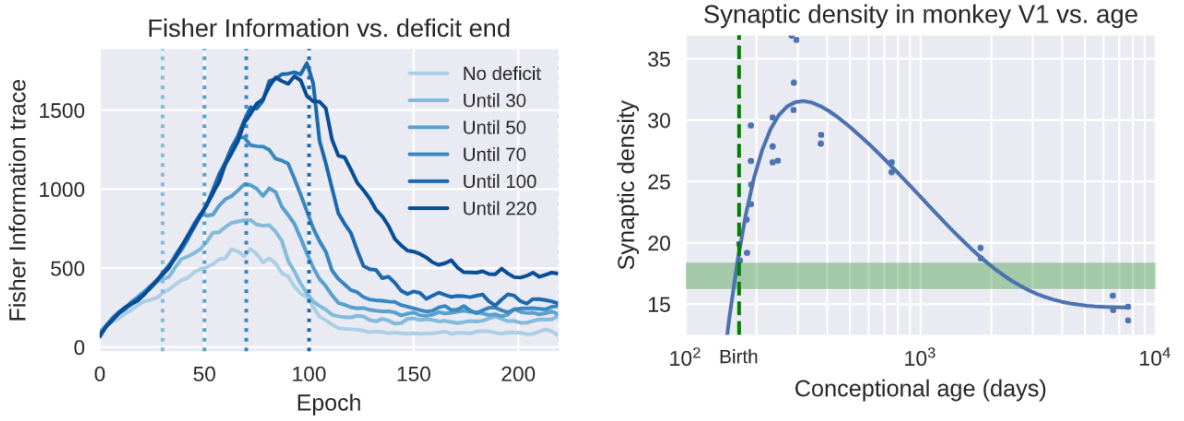


Figure 2.4: The blue line illustrates how classification accuracy on the test set degrades in relation to the duration of pretraining on the deficit-induced dataset. The green line shows the dynamics of accuracy on the test set over time. The chart on the right presents the vision acuity measurements in kittens, mapped against the timeline of when the deficit was rectified.

is too large to compute, so the authors utilize its trace to measure global or layer-specific connection strength. The authors indicate that while initially the network accumulates information about the data (leading to an increase in connection strength), after reaching a certain level of performance, the connection strength begins to decrease. This phenomenon is likened to "forgetting" or "compression", during which redundant connections are eliminated. A similar process occurs in the brain, initially characterized by the intense formation of new synaptic connections, followed by the elimination of certain connections and the strengthening of others. This process is crucial for maturation and optimization of brain functions, allowing the removal of excessive or unnecessary connections and enabling the brain to operate in a more efficient and effective manner. Figure 2.5b presents the course of such a process in macaques.

The researchers point out that these studies link biological and computational learning approaches. They suggest that critical periods may not be exclusive to biological systems but could also emerge in general learning systems, influenced by inherent constraints stemming from the dynamics of learning and information processing.

Fisher information is connected to the curvature of the training loss landscape, thereby linking it to the optimization process. As a result, the TFIM analysis also provides insights into the geometry of the loss function and the learning dynamics. The early stage of training is crucial in guiding the network towards the appropriate final 'valley'. The critical period concludes once the network overcomes all bottlenecks in the loss function landscape.



(a) Trace of Fisher Information Matrix curves for various durations of pretraining on data induced with a deficit (blurred), demonstrating the network’s adaptive information processing in response to sensory deficits.

(b) Synaptic density in monkey V1 as a function of age, illustrating synaptic pruning during neural development.

Figure 2.5: The plots taken directly from the discussed paper Achille et al. 2019 present measurements of connections in both biological and artificial information systems.

2.3. Critical Learning Periods for Multisensory Integration in Deep Networks (Kleinman et al. 2022)

Another study addresses the ability of DNNs to integrate information from diverse sources. Researchers emphasize that exposure to correlated signals during the initial training phases is crucial for the effective integration of information from two distinct sources, and introducing disturbances during these critical periods can permanently impair the DNN’s ability to utilize synergistic information.

In this study, paralleling the Achille et al. 2019 study, researchers investigated how pretraining the model on deficit-induced data in phase 1 affects its asymptotic performance in phase 2. In the mentioned study, the DNNs studied were unimodal, but given the research objective, selected models were expanded to a bimodal case, leading the researchers to reference neuroscientific studies on the effects of the visual system’s adaptation to the environment during a deficit, such as the works of Wiesel and Hubel on monocular deprivation, which were mentioned at the beginning of the chapter. Each model’s structure was modified to include two separate branches. The outputs of these branches were then combined additively using a late fusion technique (Joze et al. 2020). The result of this operation was then passed to the another branch, which further processed the signal. At the end of this branch, there was a layer responsible for classifying the image into the appropriate category. For the inputs to these branches, each image was split vertically into two equal halves. To mimic the deficit from monocular deprivation studies, the authors blurred the right part of the image according to the procedure described in Achille et al. 2019, while leaving the left side unchanged. This approach ensured that both sources in the experiment were symmetrical and homogeneous, ensuring that any asymmetries result from the discussed input corruption and not from inherent differences in the data. Similar to the Achille et al. 2019 study, the bimodal DNN showed lower classification efficiency at the end of the second phase, which was more pronounced the

longer the first phase lasted, what can be observed in Figure 2.6. However, this decline was not as pronounced as in the unimodal network experiment, because in the case of a bimodal network, when sources are symmetrical in terms of information, one branch can provide sufficient information to significantly improve classification accuracy. The authors draw parallels to the visual system studies, noting that individuals with conditions like strabismus or amblyopia often perform just as well on most tasks, like typical individuals, since the separate information coming from each sensor is often sufficient to compensate. Therefore, reporting only the model’s accuracy in the second phase might not reflect the effects of how the deficit influences how the model is organized internally.

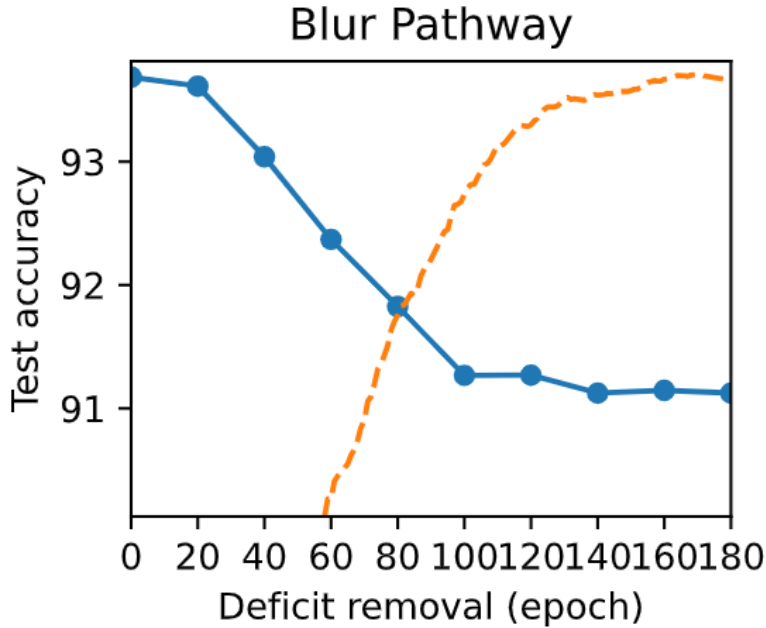


Figure 2.6: Similar to Figure 2.4, the blue line illustrates the degradation of classification accuracy on the test set in relation to the duration of pretraining on the deficit-induced dataset. The orange line depicts the dynamics of accuracy on the test set over time. In this case, the degradation of classification accuracy is not as severe as in the unimodal network, because the branch processing the uncorrupted part of the image is sufficient for partial compensation.

The Relative Source Variance (RSV) measure was introduced to quantify each source’s contribution to individual neurons’ responses, producing results within the range of -1 to 1. A result of -1 indicates that only the left source contributes to the neuron’s response, whereas a result of 1 signifies exclusive contribution from the right source. A score of zero implies that the neuron is equally sensitive to both sources. The results of this measure for all neurons of a specific layer and network inputs were collected and presented as a histogram, which assesses the current state of the network’s responsiveness to each source. In this way, the authors tracked how the RSV distribution changed in response to sensory deficits, disturbances, and during regular training. This measure is analogous to the one used for obtaining information about neuronal activity in response to stimuli delivered to each eye in the Wiesel and Hubel study. Analysis using the RSV measure revealed that blurring the right sensor during the initial learning period had a lasting effect on the network’s utilization of information from

that sensor, persisting even after the deficit was rectified. By analyzing the RSV measure, it can be observed that the network units evenly attend to both sensors after training in which the first phase lasted 0 epochs. However, prolonged pretraining on deficit-induced data led to neurons predominantly encoding information from the sensor that was not affected by the deficit in the first phase. This is illustrated by the concentration of RSV units around the value of -1, as can be observed in Figure 4.6. This behavior is analogous to the ocular dominance observed in monkeys with cataracts, as described in Figure 2.2.

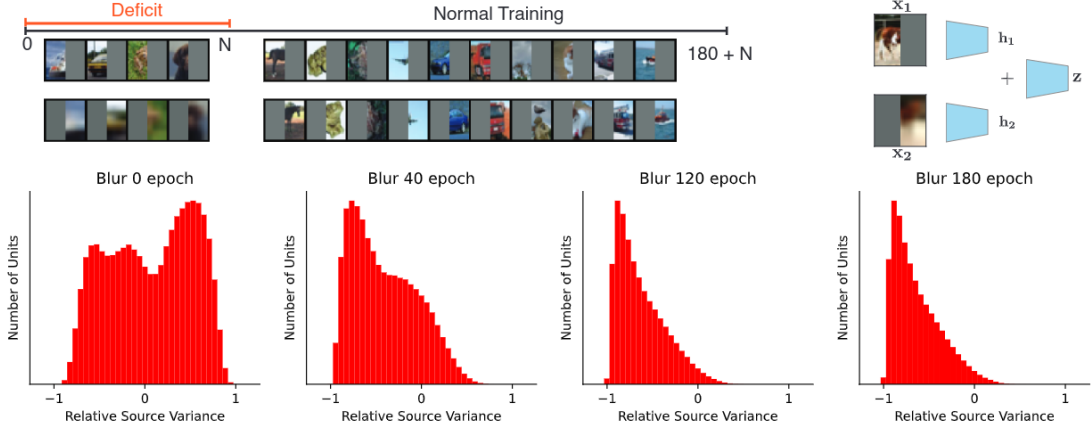


Figure 2.7: The upper part of the image illustrates the training process of a bimodal DNN, showing the division into learning phases and data processing. The first phase includes training on data where one part of the image has been blurred. The second phase involves normal subsequent training. A single image, split into halves, is used as the input for the network, with each half being fed to a different branch. The lower part of the image shows the distribution of the RSV measure at the end of the second phase, depending on the epoch at which the first phase ended. We can observe that prolonged pretraining on deficit-induced data resulted in neurons predominantly encoding information from the sensor that was not affected by the deficit in the first phase.

The important assertions for this work are the researchers' hypotheses:

1. The analysis of learning dynamics during DNN convergence is irrelevant for signal integration and the ability to extract synergistic information, as they depend solely on what happens during the initial transient learning.
2. Features from sources inhibit each other because they are competing to solve the task.

Both of them will be undermined later in the work.

2.4. Correction of amblyopia in cats and mice after the critical period (Fong et al. 2021)

Traditional amblyopia treatment methods primarily aimed at correcting the sensory organ's functionality, followed by temporarily covering the healthier eye (using the occlusion method) to encourage the use of the amblyopic eye - the eye that did not develop correctly during the monocular deprivation (MD) period. However, this approach's effectiveness is notably limited, particularly when initiated after a specific 'critical period' in childhood, with about

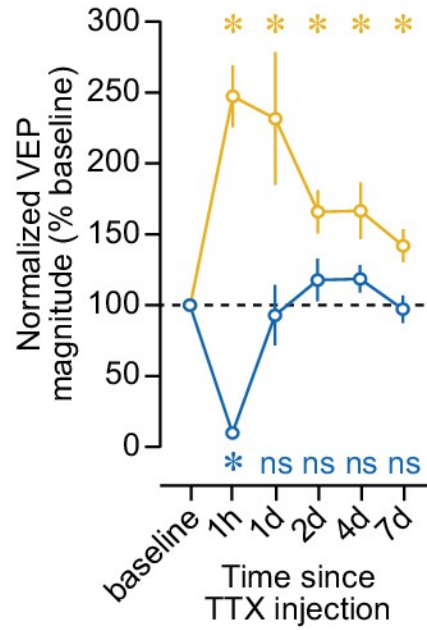
a quarter of patients experiencing vision deterioration in the amblyopic eye within a year of completing treatment. This study was motivated by numerous clinical reports indicating that stable and severe amblyopia could diminish in adults following damage or removal of the healthy (fellow) eye. The experiments in this study aim to test the hypothesis that remission is possible by eliminating activity in fellow eye just long enough for the connections from the amblyopic eye to be re-established in V1. Tetrodotoxin (TTX) was employed for the temporary deactivate the retina of the eye, with activation levels being measured by baseline visual evoked potentials via electrodes implanted in V1 layer 4.

Animal experiments (on mice and cats) showed:

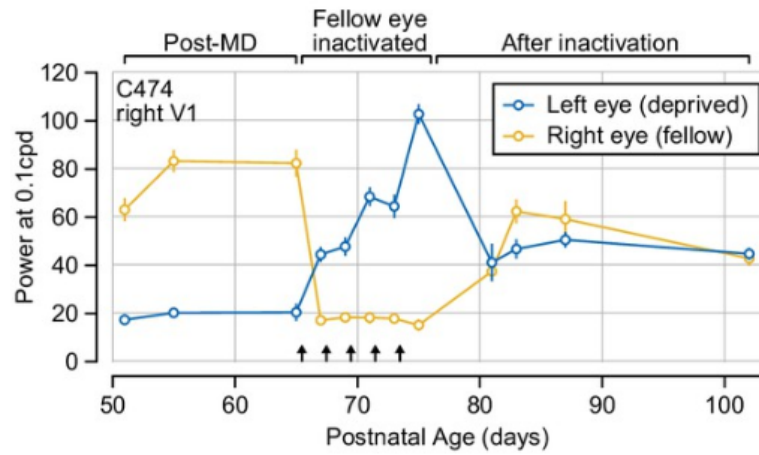
1. In typical individuals, temporarily deactivating the fellow eye results in increased activity in the non-deactivated eye. This heightened response persists at a level above baseline even a week after the deactivation has ceased. Meanwhile, the responses from the dominant eye revert to baseline levels. This dynamic is illustrated in Figure 2.8a.
2. The temporary deactivation of the fellow eye is sufficient to achieve a complete and lasting cure of amblyopia, even at ages where traditional methods are ineffective. This process exhibits dynamics similar to those described in point 1, as illustrated in Figure 2.8b.
3. The plasticity induced by refractive occlusion therapy in juvenile mice is temporary, whereas regeneration driven by the inactivation of the fellow eye is permanent. Amblyopia recurrence can occur in former case.
4. Cats subjected solely to monocular deprivation (MD) exhibited the typical reduction in cell size within the deprived layers of the dorsal lateral geniculate nucleus (dLGN) compared to the non-deprived layers. In contrast, cats that experienced inactivation of the fellow eye after MD displayed similar cell sizes in both the deprived and non-deprived layers of the dLGN. Moreover, this inactivation effectively reversed both the physiological and anatomical impacts of MD on the visual pathway in cats.

The article postulates two hypotheses. The first, "removal of suppression," suggests that amblyopia is immediately alleviated by disabling the healthier eye, but only for the duration of its deactivation. The second, "synaptic strengthening" driven by a pursuit of homeostasis, suggests that the ability to see will gradually improve and the improvement will persist beyond the deactivation period. The authors' experiments supported the latter hypothesis, demonstrating that temporarily deactivating the healthier eye after prolonged monocular deprivation can lead to a permanent correction of vision in both eyes. This outcome is thought to arise because inactivating the fellow eye establishes conditions within the visual cortex that enable the previously weak connections from the amblyopic eye to gain strength.

Reflecting on their findings, the researchers note that while studies over the past four decades have shown that synaptic connections in the primary visual cortex remain plastic throughout life, leveraging this plasticity to benefit people has been challenging. However, they now suggest that connections serving the amblyopic eye are capable of substantial plasticity beyond the critical period, and this potential is unleashed by reversibly silencing the fellow eye.



(a) This graph illustrates the significant increase in normalized visual evoked potential (VEP) magnitude within the first hour post-injection of TTX, followed by a decline towards baseline levels over the course of seven days, in neurotypical mice, initiated at postnatal day (P) 47 which is after the critical period has ended.



(b) The graph displays the visual cortex response in the right (fellow) and left (deprived) eyes of a cat post-monocular deprivation (MD) and following fellow eye inactivation. The response of the deprived eye peaks after fellow eye inactivation and then gradually declines, reaching the activation level of the fellow eye.

Figure 2.8: Activation levels of both eyes in individuals recorded after the critical development period and the impact of the applied intervention on these activation levels. At the top, a neurotypical individual; at the bottom, an individual subjected to monocular deprivation. For a detailed description of the symbols, please refer to Fong et al. 2021.

Chapter 3

Description of the experiments

This chapter begins with a general overview, then progresses to a detailed description of the experiments. The outcomes of these experiments will be presented in the following chapter.

3.1. General description

In the following experiments, an intervention will be conducted on a bimodal DNN, inspired by the deactivation of the fellow eye in the visual system, which inspiration is discussed in Section 2.4 in Chapter Two. It was chosen due to the simplicity of implementing "deactivation" of one branch of bimodal DNNs. The analysis of inspiration using the occlusion method is left for further research.

It's important to note that the experiments conducted in this study do not aim to directly model the phenomena observed in the visual system. Instead, their purpose is to investigate a method of correcting phenomena identified in the first phase of the aforementioned study of Kleinman et al. 2022, which is inspired by research on the proposed correction of amblyopia after the critical period in animals. At the same time, the author of this work is aware that amblyopia is a deficit related to the eyes, not the visual fields. Therefore, the discussion here revolves around inspiration from phenomena in the visual system, rather than pointing to direct correlations between the biological and artificial systems. When discussing about equivalents in the two given systems, this should be taken into account.

As a disturbance leading to the equivalent of amblyopia in DNNs, a counterpart to cataract (blurring) was chosen, as in the study Kleinman et al. 2022.

The ResNet-18 architecture [He et al. 2016] was modified to process bimodal input. This was achieved by adding a duplicate of the initial four blocks of the ResNet-18 network, resulting in a model comprising two branches – referred to as the left and right branches for clarity – that process input data and generate outputs. These outputs are then combined using the late fusion method [Joze et al. 2020] through an addition operation. The result of this operation is passed to the main branch, which corresponds to the remaining part of the ResNet-18 network, for further signal processing. At the end of the main branch is located a classification layer responsible for determining, through a probability distribution generated by the softmax function, the class to which the input belongs. This architecture is designated as S-ResNet-18, in line with Kleinman et al. 2022.

For this study, the CIFAR10 dataset [Krizhevsky & Hinton 2009] is adapted for bimodal classification task such that each image is vertically split into two equal parts. The resulting dataset is dubbed S-CIFAR10. Additionally, the name SBR-CIFAR10 is used to denote the dataset where the right half of each image undergoes a four-fold reduction in resolution (blurring) before being restored to its original resolution.

In training the selected model on these datasets, the cross-entropy loss function is employed, while the optimization is carried out using Stochastic Gradient Descent (SGD) [Bottou 1998].

During the intervention phase, the chosen deficit is applied only to the part of the image that serves as input to the right branch of the bimodal DNN, while the left branch undergoes deactivation.

3.1.1. Prestudy: Baselines

The study focused on evaluating the performance of the bimodal model when it has access to only one visual field, achieved by deactivating one of its branches.

- The left branch was deactivated, and the right visual field without deficit was fed to the right branch.
- The right branch was deactivated, and the left visual field without deficit was fed to the left branch.
- The left branch was deactivated, and the right visual field with an induced deficit was fed to the right branch.

The accuracy results obtained at the end of the epoch should be considered as a baseline, which should not be exceeded in any subsequent trainings. This is because it would suggest that the additional information or the action taken is detrimental to the training.

The classification results can be observed as dashed lines, for the first two cases, in Figure 4.3a and Figures 4.1. The third classification accuracy result was omitted from the graphs as it obscured the clarity of the graphs due to its significantly lower value compared to the others. This result was a classification accuracy of **60.87 %** on the test set.

3.1.2. Main study

The main study was divided into four phases:

1. **Phase One (Pretraining with deficit):** The selected model undergoes training on the chosen dataset, where a portion of the images fed into to the right branch are subjected with a chosen deficit.
2. **Phase Two:** The model continues training on deficit-free data for a specified number of epochs.
3. **Phase Three (Intervention):** The intervention is applied to the left branch of the model.
4. **Phase Four:** The model then resumes training, following the same procedure as in the second phase, until it reaches the designated accuracy level on the training set.

In the main study, the number of epochs for phases one and three are variables, used to investigate the effect of subjecting the model to a deficit on subsequent training, and the effect of applying an intervention on a model after a specific duration of the first phase, respectively.

Phase one can last for $x_1 \in \{0, 40, 80, 120, 160, 200\}$ epochs.

Phase two lasts for $x_2 = 200$ epochs.

Phase three can last for $x_3 \in \{0, 20, 40, 60, 80, 120, 160, 200\}$ epochs.

The duration for Phase Four is $x_4 =$ when the classification accuracy on the training set reaches the final classification accuracy after 200 epochs for the training where phase 3 lasted 0 epochs.

As evident from Figure 4.1, results are only comparable within the scope of a fixed value of x_1 . Therefore, compensating for the length of the second phase, in the fourth phase, to equalize training isn't significant here, as long as we aren't comparing results between outcomes of the first phase.

In this study, the final image classification accuracy in the second phase is used to investigate the effect of the deficit's duration on subsequent training. In contrast, the final image classification accuracy in the fourth phase is used to study how the length of the applied intervention impacts the model's performance when trained on deficit-induced data.

A discussion of the results of this experiment can be found in Section 4.1.

3.1.3. First Substudy: Effective plasticity

The next study involves an analysis using the Fisher Information Matrix trace $\text{Tr}(\mathbf{F})$, to assess the network's activation levels. According to the interpretation presenting synaptic strength of a connection between neurons in DNNs, as derived in Achille et al. 2019 and discussed in Section 2.3, the Fisher Information Matrix ($\mathbf{F}$) can be utilized to measure the contribution of a selected weight to the network's output when that weight undergoes perturbation. Given the impracticality of computing $\mathbf{F}$ for the deep networks considered in this study on the available GPU, an approximation of the Fisher Information Matrix trace will be calculated as follows:

$$\text{Tr}(\mathbf{F}) = \mathbb{E}_{x \sim Q_X, \hat{y} \sim p_\theta(y|x)} \left[\left\| \frac{\partial \log p_\theta(\hat{y}|x)}{\partial \theta} \right\|_2^2 \right] \approx \frac{1}{N} \sum_{n=1}^N \frac{1}{M} \sum_{m=1}^M \left\| \frac{\partial \log p_\theta(\hat{y}_{mn}|x_n)}{\partial \theta} \right\|_2^2 \quad (3.1)$$

following Achille et al. 2019 and Jastrzebski et al. 2021.

This study is also inspired by the research in Fong et al. 2021, which examined activation levels using Visual Evoked Potentials, as discussed in Section 2.4. Similarly, in the current study, the activation dynamics of each branch of the examined model will be investigated, depending on its current phase.

In Section 4.2, the impact of the S-CIFAR10 and SBR-CIFAR10 datasets on the balance of $\text{Tr}(\mathbf{F})$ between the branches of the S-ResNet-18 network, and correlational studies between the aforementioned balance, in the final training phase, and the final classification accuracy are introduced.

For brevity in the discussion, the notations $\text{Tr}(\mathbf{F}_l)$ and $\text{Tr}(\mathbf{F}_r)$ will denote the $\text{Tr}(\mathbf{F})$ values for the left and right branches of the studied model, respectively.

3.1.4. Second Substudy: RSV

The main study will also be accompanied by a measurement of the Relative Source Variance (RSV), as defined in 1.0.11, during phase two. This measurement will indicate the dependency of unit activations in a selected layer of the main branch on the input source, based on the duration of phase one. This approach aligns with the methodology used in the study conducted in Kleinman et al. 2022, as described in Section 2.3. Additionally, this study will explore the activity of units in the chosen layer during phase four, examining how this activity varies with the length of the intervention.

RSV was applied to the first layer of the shared branch.

The results of the experiment are presented in Section 4.3.

3.1.5. Third Substudy: Impact of Learning Rate and Weight Decay on Imbalances in Branch Reliance

The study examined the impact of selected hyperparameter sizes on the ratio $\frac{\text{Tr}(\mathbf{F})_l}{\text{Tr}(\mathbf{F})_r}$, the significance of which was elaborated upon in Section 4.2. The chosen hyperparameters are the learning rate and weight decay in the SGD optimizer. Additionally, the final test classification accuracy was provided for the dynamics of the aforementioned ratio.

The results of this study are discussed in Section 4.4.

3.1.6. Additional Substudy to Main Study: Another Models and Datasets

In conjunction with the main study, an auxiliary investigation was conducted using a different DNN trained on a different dataset, as well as a modified version of the main study, specifically in the setting where an intervention (phase 3) follows the deprivation phase (phase 1).

The results of these experiments are presented in Section 4.5.

3.1.6.1. S-SimpleCNN without skip connections on SBR-FMNIST and S-FMNIST

This model, distinct from the S-ResNet-18, does not incorporate skip connections and is intentionally overparameterized, containing approximately 21 million parameters. This design choice was made to challenge the model’s capabilities in relation to the dataset’s size. This model will be referred to as S-SimpleCNN. Datasets derived from Fashion MNIST [Xiao et al. 2017], modified for the bimodal task analogous to the process described for the CIFAR10 dataset, are dubbed SBR-FMNIST and S-FMNIST.

Model Architecture The model is structured similarly to the S-ResNet-18 network, featuring two input-processing branches that converge into a common branch. Each input branch comprises a sequence of convolutional [LeCun et al. 1998] and batch normalization [Ioffe & Szegedy 2015] layers, organized into specific blocks as follows:

- The initial block in each branch consists of:
 - Conv2d layers with configurations: Conv2d(32, 3, 1, 1), Conv2d(32, 3, 1, 1), and Conv2d(32, 5, 2, 2). The parameters for these layers represent the number of filter outputs, kernel size, stride size, and padding size, respectively.

- BatchNorm2d layers following each Conv2d layer.
- ReLU activation function applied after each BatchNorm2d layer.
- This is followed by three repeated blocks, each containing:
 - Conv2d layers configured as: Conv2d(256, 3, 1, 1), Conv2d(256, 3, 1, 1), and Conv2d(256, 5, 2, 2).
 - BatchNorm2d and ReLU layers, similar to the initial block.

Main Branch Configuration The main branch of the network, where the outputs from the two input branches are integrated, consists of:

- Three blocks, each with the following layers:
 - Conv2d(256, 3, 1, 1), BatchNorm2d, and ReLU layers, repeated three times within each block.
- A linear transformation layer: Linear(512, 128), followed by BatchNorm1d and ReLU.
- A final linear layer: Linear(128, 10), which outputs the classification results.

This architecture aims to effectively process and integrate the bimodal input data, leveraging the depth and complexity of the network to achieve accurate classification on the Fashion MNIST dataset.

3.1.6.2. S-ResNet-18 on CIFAR10 with Altered Phase 4 Setting

In the main study, the classification accuracy of the S-ResNet-18 model on the S-CIFAR10 dataset did not reach the level presented in the article Kleinman et al. 2022, the results of which are shown in Figure 2.6. Therefore, the setting of the fourth phase of the experiment is modified to achieve comparable classification performance, in the best case. This modification was also intended to test the effectiveness of the applied intervention in this new setting.

3.1.7. Additional information

Image Preprocessing:

- The selected image undergoes a horizontal flip operation with a probability of 0.5.
- The image is divided, vertically, into two equal parts.
- A fourfold reduction in resolution is applied, followed by a return to the original resolution - for the right half of the image in the first phase.
- Each half is subjected to a random shift of 1/8 of the image size, both horizontally and vertically.
- The halves are normalized according to the parameters estimated for the entire training set - for each half separately.

3.2. Detailed description

3.2.1. Main study

In the main study, the parameter space was explored as follows:

- Parameters were explored based on the best accuracy on the test set, for $lr \in \{1e-4, 1e-3, 1e-2, 1e-1, 1e-0\}$ within a budget of 200 epochs for the S-CIFAR10.
- The learning rate was then explored in the set $lr \in \{2e-x, 4e-x, 6e-x, 8e-x\}$, where x is the order for which $lr = 1e-x$ returned the best accuracy, for training duration matching the previous one.
- The search was conducted separately for the data used in phase 1 and the data used in phase 2.
- Each search was conducted with a single seed.

In the additional experiments related to the main task, the procedure is identical to the main experiment, with the only differences being the budget of epochs and the choice of dataset.

In all DNNs discussed in this work, convolutional layers' weights are initialized with Kaiming Normal initialization [He et al. 2015] in fan-out mode. Batch normalization layers, have their weights set to ones and biases to zeros. For fully connected layers, weights are uniformly initialized within a range determined by the inverse square root of the number of output features, and biases are initialized to zero.

3.2.1.1. Research Details: Main Study when phase 2 lasted 0 epochs

In the research, the S-ResNet-18 model was utilized, with the SBR-CIFAR10 dataset for the first phase and S-CIFAR10 for subsequent phases. The chosen criterion was Cross Entropy Loss, and SGD served as the optimizer. The experiment's hyperparameters featured a variation in epochs, with x1 for the first phase, 0 for the second, x3 for the third, and x4 for the fourth phase, along with a consistent batch size of 125, a learning rate of 6e-1, and an absence of weight decay throughout.

3.2.1.2. Research Details: Main Study when phase 2 lasted 200 epochs

In this phase, the S-ResNet-18 model was again employed, initially using the SBR-CIFAR10 dataset and then shifting to S-CIFAR10. The criterion and optimizer remained consistent with the previous phase. The hyperparameters for this phase featured variations in epochs, set as x1 in the first phase, 200 in the second, x3 in the third, and x4 in the fourth, along with a steady batch size of 125, a learning rate of 6e-1, and a constant weight decay of 0.0.

3.3. Prestudy: Baselines

For all cases in the prestudy, the following settings were employed:

3.2.2.1. Research Details: Baselines

The model used was S-ResNet-18 with datasets SBR-CIFAR10 or S-CIFAR10. Cross Entropy Loss was the criterion, and SGD was the optimizer. The hyperparameters were a fixed 200 epochs, batch size of 125, learning rate of 6e-1, and weight decay at 0.0.

3.3.1. First Substudy: Effective plasticity

The $\text{Tr}(\mathbf{F})$ was computed every $\frac{1}{2}$ epoch, using held-out 2% of data from the training set. The data was sampled such that the class distribution was uniform.

In all cases, $\text{Tr}(\mathbf{F})$ is computed using held-out data from the same dataset on which the training is currently conducted, while excluding weights that are parameters of the batch normalization layers.

All the charts presenting these values have been smoothed using the Exponential Weighted Moving Average over the last 10 steps.

It should be noted that in the studies in Section 4.2, the same hyperparameters as in Study 3.2.1 were used.

3.3.2. Second Substudy: RSV

- From the training set, $c \cdot 5$ images were selected (5 per class), where $c = 10$ represents the number of labels in the dataset.
- For each image, and for each of its halves, 99 augmentations were performed using random translations up to 4 pixels and random rotations up to 10 degrees. This procedure was employed to obtain a distribution of elements from the source. The subsequent values calculated, as described by definition 1.0.11, were compiled into a histogram. As a result, we obtained $c \cdot 5 \cdot 200$ data points.
- Measurements were taken at selected checkpoints at the end of phase 2 and 4.

On the histogram charts, both the median and the mean are provided to facilitate the comparison of plots. The charts lack confidence intervals because, with the chosen sample of 576,000 values, their lengths are less than $5e-3$ in each case, which on the scale of the entire interval is indistinguishable from the centers of these intervals.

3.3.3. Third Substudy: Impact of Learning Rate and Weight Decay on Imbalances in Branch Reliance

The selected hyperparameters, learning rate (lr) and weight decay (wd), belong to the following sets — this is an exhaustive pairwise study:

$$lr \in \{5e-3, 5e-2, 5e-1\}$$

$$wd \in \{0.0, 1e-4, 1e-3, 1e-2\}$$

Lower orders of learning rate than those listed lead to poorly convergent trainings.

The L2 regularization excluded the parameters of the batch normalization layers.

3.3.4. Additional Substudy to Main Study: Another Models and Datasets

Similar to the main study, the number of epochs for phases one and three are variables, used to investigate the effect of subjecting the model to a deficit on subsequent training, and the effect of applying an intervention on a model after a specific duration of the first phase, respectively.

3.2.6.1. Research Details: S-SimpleCNN on the Fashion MNIST when phase 2 lasted 0 epochs

Phase one can last for $x_1 \in \{0, 20, 40, 60, 80, 120\}$ epochs.

Phase three can last for $x_3 \in \{0, 20, 40, 60, 80, 120, 160\}$ epochs.

The duration for Phase Four is $x_4 =$ when the classification accuracy on the training set reaches the final classification accuracy after 300 epochs for the training where phase 3 lasted 0 epochs.

The study utilized the S-SimpleCNN model on two datasets: SBR-FMNIST for phase one and S-FMNIST for subsequent phases. The criterion used was Cross Entropy Loss with SGD as the optimizer. In the experiment, key hyperparameters were varied across different phases: the epochs were set as x_1 for the first phase, 0 for the second, x_3 for the third, and x_4 for the fourth. Additionally, the batch size was consistently 125, the learning rate was maintained at $1e-1$, and the weight decay was uniformly set at 0.0 throughout the experiment.

3.2.6.2. Research Details: S-ResNet-18 on CIFAR10 with Altered Phase 4 Setting when phase 2 lasted 0 epochs

The study was modified in comparison to the main study.

Phase one can last for $x_1 \in \{0, 20, 40, 60, 80, 120, 160, 200\}$ epochs.

Phase three can last for $x_3 \in \{0, 20, 40, 60, 80, 120, 160\}$ epochs.

In this part of the study, the S-ResNet-18 model was employed, working with SBR-CIFAR10 in phase one and S-CIFAR10 in later phases. The chosen criterion was Cross Entropy Loss, and SGD was used as the optimizer. A distinct feature of this experiment involved the implementation of a Multiplicative Scheduler, characterized by a sequence of values: initially 1.0 for the first three phases and then 0.98 for the fourth phase. In this setup, the learning rate is adjusted at the end of each epoch by being multiplied by these specified values, corresponding to their respective phases. The hyperparameters for this experiment included a range of epochs, represented as x_1 in the first phase, 0 in the second, x_3 in the third, and x_4 in the fourth. The batch size was consistently set at 125, with a learning rate of $6e-1$. The weight decay was varied across phases, being 0.0 in the first three phases and then adjusted to $5e-4$ in the fourth phase.

Chapter 4

Experiments

This chapter summarizes the experimental results. Section 4.1 reveals that the intervention (phase three), involving the deactivation of one of the branches for a specified number of epochs, enables the bimodal S-ResNet-18 network to correct the final classification accuracy after training on the uncorrupted dataset (phase four), on the bimodal test dataset S-CIFAR10. This holds true regardless of initial training duration on the SBR-CIFAR10 dataset (phase one) - where each S-CIFAR10 input is partially blurred - and irrespective of whether, between phase one and the intervention phase, the network had access to the unaltered S-CIFAR10 dataset (phase two).

In Section 4.2, an explanation is attempted for the phenomenon of classification performance degradation observed at the end of the second phase, as the training lengthens in the first phase. This explanation is based on the trace of the Fisher information matrix $\text{Tr}(\mathbf{F})$, which quantifies the aggregated impact of a model’s weights on its output at a given training step. This topic was discussed in Section 2.2 and Subsection 3.1.3. This measure indicates that when the network is pretrained on uncorrupted data, both branches of the network contribute similarly to the model’s response. However, when pretrained on the corrupted dataset, the weights of the branch processing the corrupted image part (right branch) has a markedly lower impact than weights of the branch (left branch) processing the uncorrupted part. This section also examines how the balance in the weight contributions of both branches correlates with the final classification accuracy on the test dataset.

In Section 4.3, the study introduces a novel measure to determine the model’s dependency on its branches. This measure evaluates the sensitivity of each neuron in the shallowest layer of the common branch to input perturbations, specifically at the conclusion of training on the S-CIFAR10 dataset. The sensitivities are then compiled into a histogram. The results suggest that extended training during phase one leads to an increased sensitivity to disturbances in inputs to the left branch by the end of phase two. Interestingly, a prolonged intervention period (phase three) rectifies this imbalance in sensitivity by phase four, mirroring the scenario observed at the conclusion of phase two when the network had not undergone training on the corrupted dataset.

Section 4.4 investigates how hyperparameters like weight decay and learning rate impact branch weight reliance during training on SBR-CIFAR10. Higher values in both hyperparameters lead to more pronounced branch imbalances but don’t necessarily degrade final test performance. The section posits that the correlation between final performance and branch

weight balance observed in Section 4.2 is consistent within the established hyperparameters.

The final Section 4.5 presents studies that extend the main study from Section 4.1.

4.1. Main study

Drawing inspiration from the aforementioned study Fong et al. 2021, in the following experiment, the main study from Kleinman et al. 2022 is extended by adding a third and fourth phase, as described in Section 3.1.2, when training the S-ResNet-18 model on the S-CIFAR10 dataset.

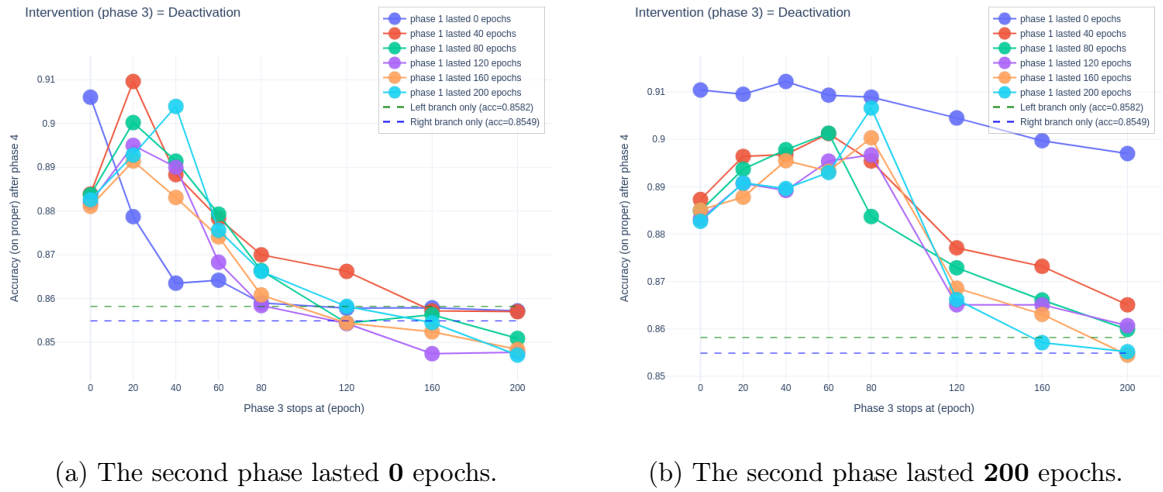


Figure 4.1: The impact of intervention duration on the test classification accuracy of the S-ResNet-18 model, at the end of phase 4, on the S-CIFAR10 dataset is analyzed. From both plots, it can be inferred that in every scenario where the DNN had access to corrupted data (SBR-CIFAR10), a certain duration of intervention allowed the network to correct its classification accuracy. However, excessively long interventions could lead to diminished performance. The only situation where intervention consistently has a negative impact is when the network did not have access to corrupted data. Nevertheless, if the network, in this case, is pretrained on the uncorrupted dataset (S-CIFAR10) for **200** epochs, the negative impact of the intervention is less pronounced.

Figure 4.1 shows the impact of intervention duration on the final model performance in two cases - when the network did not learn to integrate two correct halves of the image in the second phase of the experiment, and when it learned this integration over **200** epochs. In each of the two cases, for each non-zero duration of training the model on deficit-induced data, there exists an epoch up to which every available intervention duration will lead the model towards better final performance in the fourth phase compared to not intervening at all. However, intervention durations longer than these epochs not only fare worse than not intervening, but they may also perform worse than the established baselines of models trained solely on one of the branches.

It is worth drawing a parallel with the studies in Fong et al. 2021, where, as mentioned in the related works Section 2.4, for typical subjects, applying an intervention after the critical period does not lead to drastic changes. Similarly, in the scenario where phase 1 lasted **0**

epochs, a short intervention (phase3) duration does not drastically affect classification accuracy for situation where phase 2 lasted **200** epochs, in contrast to the same intervention duration when phase 2 lasted **0** epochs.

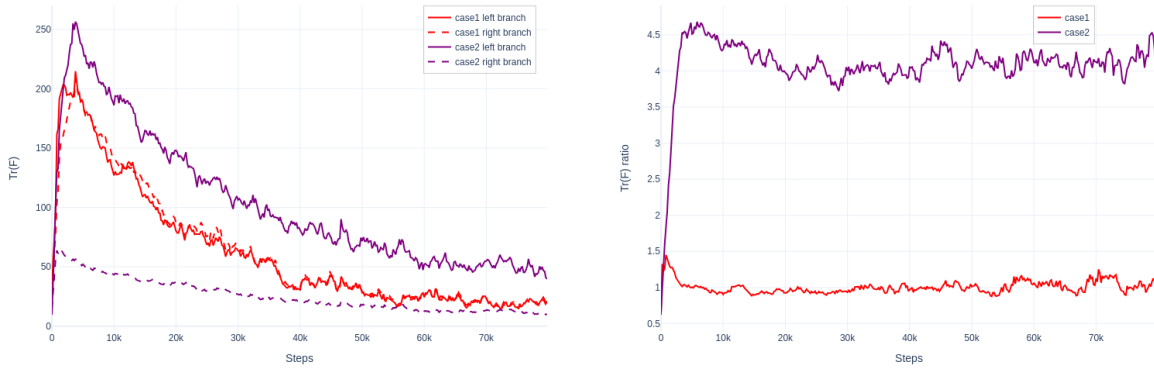
Looking closely at Figure 4.1a for the case where the third phase lasted **0** epochs (then phase 4 can be seen as phase 2), we can observe the impact of pretraining length on deficit-induced data on further model training - even **40** epochs of pretraining on corrupted data is enough for the model to significantly decrease its performance, grouping with the rest of the possibilities.

Discrepancies between the $\text{Tr}(\mathbf{F})$ values of both branches right from the start of training, what can be observed in Figure 4.2a, may be indicative here, and the following sections will further elaborate on it.

4.2. First Substudy: Effective plasticity

Figure 4.2 illustrates the $\text{Tr}(\mathbf{F})$ of the S-ResNet-18 model's branches in two cases:

1. Model trained on S-CIFAR10 (*red*).
2. Model trained on SBR-CIFAR10 (*purple*).



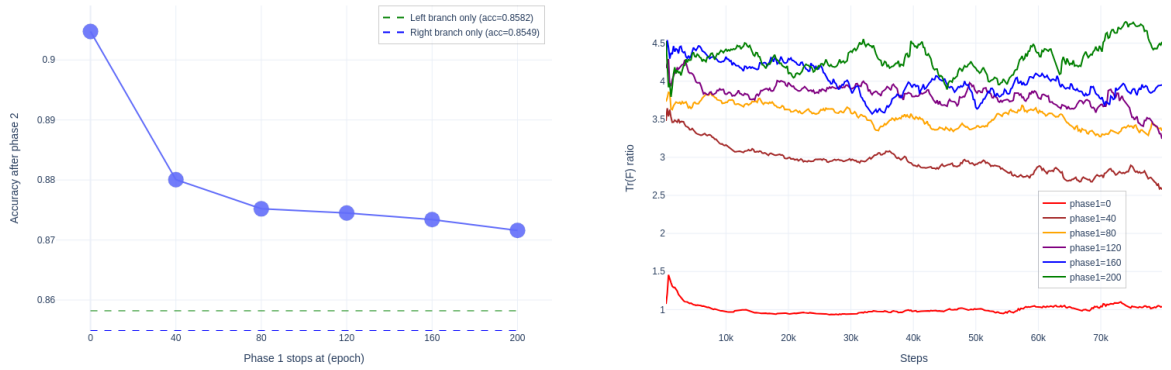
(a) The $\text{Tr}(\mathbf{F})$ for the left and right branches over the course of training. In the case of introducing a corruption (blurring the right part of an image) into the dataset, the weights of the branch processing the uncorrupted part of the image contribute more to the network's final prediction than the weights of the second branch, and more than the weights of both branches when the network is trained on the dataset without introduced corruption. Simultaneously, the weights of the branch processing the corrupted part of the image contribute much less than the weights of the other mentioned branches.

(b) The ratio of $\text{Tr}(\mathbf{F})$ s ($\frac{\text{Tr}(\mathbf{F}_l)}{\text{Tr}(\mathbf{F}_r)}$). When calculating the ratio, it can be observed that in the case of a dataset with corruption, the weights of the branch processing the uncorrupted part of the image contribute over four times more to the final output, almost throughout the entire training duration. In contrast, in the case of a dataset without corruption, the contribution of the weights of each branch is on average comparable throughout the entire training process.

Figure 4.2: Comparison of $\text{Tr}(\mathbf{F})$ s for training the S-ResNet-18 model on S-CIFAR10 and SBR-CIFAR10.

As seen in the first case in Figure 4.2a, the $\text{Tr}(\mathbf{F})$ of both branches almost overlap. However, in the second case, $\text{Tr}(\mathbf{F}_r)$ is significantly smaller than $\text{Tr}(\mathbf{F}_l)$, almost from the very beginning. Moreover, when training on S-CIFAR10, $\text{Tr}(\mathbf{F}_l)$ is smaller than $\text{Tr}(\mathbf{F}_r)$ when trained on SBR-CIFAR10, indicating that in the second case the model tries to rely more on information from the same field of view when the second field of view does not provide it with sufficient information. To better compare the magnitudes of $\text{Tr}(\mathbf{F})$ of both branches, a plot showing the ratio of $\text{Tr}(\mathbf{F}_l)$ to $\text{Tr}(\mathbf{F}_r)$ is provided. We can observe, in Figure 4.2b, that when training on S-CIFAR10, the ratio is indeed close to 1 throughout the training period, indicating that the model gives equal importance to the weights of both branches. In contrast, when training on SBR-CIFAR10, the model gives over four times greater importance to the left branch, almost throughout the entire training duration.

As it turns out, the model’s classification effectiveness depending on the length of the first phase can be correlated with the aforementioned ratio. Figure 4.3 reveals that a higher final classification accuracy corresponds to a ratio of $\text{Tr}(\mathbf{F})$ s close to the value of 1.



(a) Classification test accuracy after the end of phase 2, depending on the duration of phase 1.

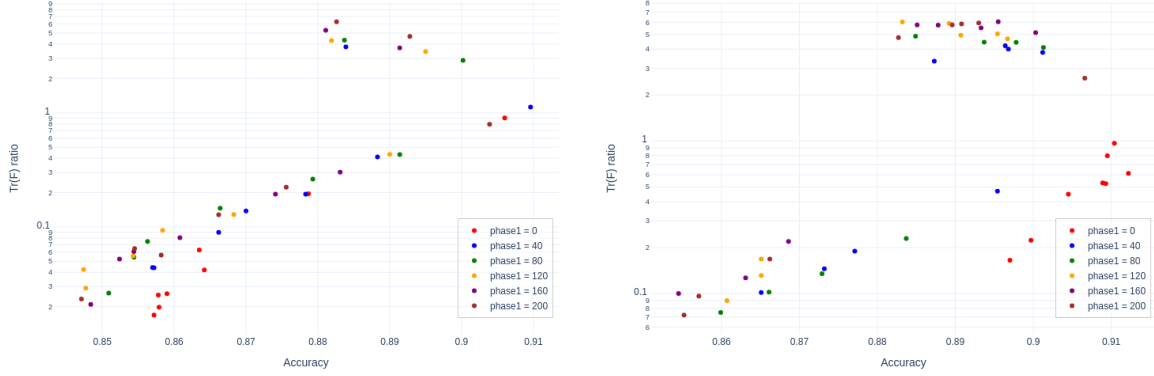
(b) The ratio of $\text{Tr}(\mathbf{F})$ s ($\frac{\text{Tr}(\mathbf{F}_l)}{\text{Tr}(\mathbf{F}_r)}$) throughout the entire training process in phase 2 for selected experiments continuing the training initiated in phase 1.

Figure 4.3: The analysis of the S-ResNet-18 model’s performance in the second phase indicates that higher final classification accuracy is associated with a $\text{Tr}(\mathbf{F})$ ratio approaching 1.

The above considerations suggest that during training on SBR-CIFAR10, the model learns to give greater importance to the weights of the left branch, while during further training on S-CIFAR10, the model learns to restore attention to the weights of the second branch, and it does so better the earlier it starts the second phase. The above results, therefore, suggest that experiencing a minor disturbance (like blurring) in the data is sufficient to alter the network’s connection structure.

Let’s examine the dynamics of the ratio of $\text{Tr}(\mathbf{F})$ s of both branches for models upon which an intervention was conducted that led to the best possible test classification accuracy in Section 4.1. To investigate the relationship between these two magnitudes, the experiment results have been compiled into Figure 4.4, a scatter plot presenting on one axis the medians of the

last **50** steps from the aforementioned ratio of $\text{Tr}(\mathbf{F})$ s and on the other axis the median of the last **10** epochs of classification accuracy for a given model. For clarity, the ratio value axis is presented on a logarithmic scale.



(a) The second phase lasted **0** epochs.

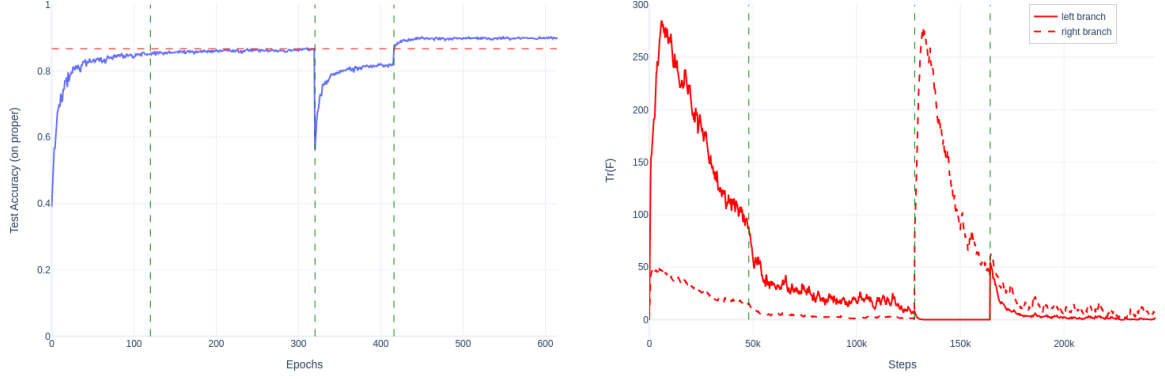
(b) The second phase lasted **200** epochs.

Figure 4.4: The relationship between the ratio of $\text{Tr}(\mathbf{F})$ s of both branches and the test classification accuracy at the end of phase 4 is evident. In both scenarios, it can be observed that regardless of the duration of the first phase, there exists a correlation between the highest achieved test accuracy and the ratio of $\text{Tr}(\mathbf{F})$ s.

As can be observed, interventions leading to the best classification accuracy cluster around a ratio of **1**, while for models with interventions leading to lower accuracy, this ratio deviates from **1**. Unfortunately, in some cases, the applied intervention did not achieve a satisfactory correction of classification accuracy. However, the presence of points on both sides of the horizontal line defined by a ratio of **1** might suggest that this is due to insufficiently dense sampling of intervention durations. This topic will be addressed in Section 4.3.

In Figure 4.5b, we can observe the progression through all training phases of $\text{Tr}(\mathbf{F}_1)$ and $\text{Tr}(\mathbf{F}_r)$, which in this study are loosely defined as the activation levels of each branch. Comparing this progression with Figure 2.8b, which displays the activation levels of neurons recorded via a non-invasive electroencephalogram over the V1 visual cortex of a cat subjected to early monocular deprivation, we can see similarities in dynamics at specific moments during the respective phases. These similarities could potentially be more pronounced if the deactivation of the network occurred gradually rather than instantaneously, as indicated by the sudden spikes observed in Figure 4.5b.

It is important to note a fundamental difference, which is the tendency of $\text{Tr}(\mathbf{F})$ to decrease as training progresses. Further literature review or studies that identify similar trends in the activity of the visual cortex are required. Nevertheless, given that changes in neuronal processing can influence brain plasticity [Wiesel 1982], and also affect evoked potentials [Purdy et al. 2001], the steps taken in this study, inspired by the methodology outlined in Fong et al. 2021, appear to be well-motivated.



(a) Test accuracy throughout the entire training process across all training phases.

(b) Trace of the Fisher Information Matrix across all training phases, calculated on the held-out data from the training set currently being used to train the model.

Figure 4.5: The progression of test accuracy and the trace of the Fisher Information Matrix throughout the entire training, encompassing all four phases, is presented. In the first phase, the S-ResNet18 model is trained on the SBR-CIFAR10 dataset for **120** epochs, where an imbalance is observed between the impacts of the weights of both branches. During the second phase, lasting **200** epochs, metric for both branches decrease, yet the disparity persists. In the third phase, one of the branches is deactivated for **96** epochs, leading to a significant activation of the other branch and a decrease in test accuracy, which had been increasing on the S-CIFAR10 dataset until this point. In the fourth phase, also lasting **200** epochs, upon reactivation of the branch, a reversal in the impact of one branch’s weights on the network’s responses is observed, along with a reduction in the disparity, resulting in improved classification accuracy on the test dataset.

4.3. Second Substudy: RSV

Researchers in Kleinman et al. 2022 investigated how the distribution of RSV of units in a specific layer, the measure that was defined in 1.0.11, changes at the end of the second phase, depending on the length of the first phase, revealing another result of interference on the network’s learning dynamics in the early stages of training, which is described in detail in Section 2.3 and which is presented in the first row of Figure 4.6.

To verify whether the introduced intervention also affects the RSV distribution after the fourth phase, thereby not only improving final test classification accuracy, additional measurements are introduced at the end of the fourth phase, for intervention durations leading to the best test classification accuracy in main study, when the second phase lasted **200** epochs 4.1b.

From Figure 4.6, we can observe that for situations where the first phase lasted **0** epochs, the RSV distribution after the second phase is close to being symmetrical. This situation changes when we allow pretraining in the first phase for at least **40** epochs, at which point the units of the selected layer begin to respond significantly more often to perturbations of the input from the left branch than to perturbations from the right branch. As the first phase lengthens, this

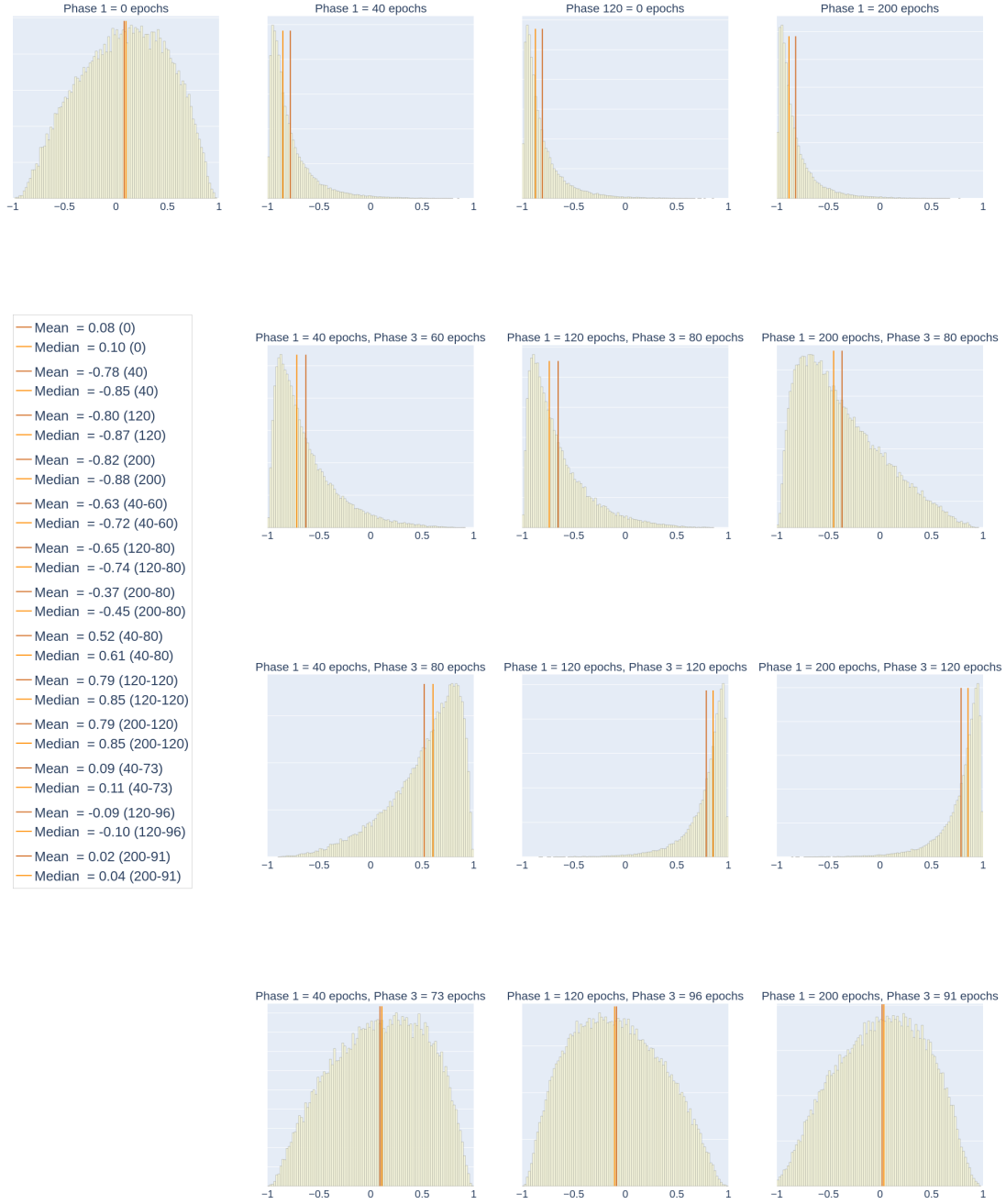


Figure 4.6: Histograms of RSV at the end of the second (top row) and fourth (bottom rows) training phases, for selected durations of the first phase (columns), are shown. The second row corresponds to the intervention results leading to the highest test accuracy in phase four, as per Figure 4.1b. The third row represents a subsequent intervention. The change in skewness of the histograms provides an indication that between these intervention durations, there is a more optimal length that will lead to a more even distribution, as demonstrated in the fourth row. The table on the left displays the median and mean for each of the histograms.

phenomenon intensifies, as can be observed through a smaller mean and median. Selecting an intervention length that yields the best final test classification accuracy in the fourth phase, from those chosen during the main experiment, also results in a model with mean and median RSV values closer to zero than before the intervention. This outcome, while not directly comparable to restoring balance between the branches, suggests something important. The direction of skewness and the contrasting skewness of histograms, particularly the RSV for models with longer interventions, hint that the lack of balance may be due to a significant difference in the chosen intervention lengths. This is supported by the bottom row of data, where models found using the bisection method for cases where the median RSV distribution deviated no more than **0.15** from the **0** value.

The improvement in accuracy due to the above-described search occurs between the second and fourth rows, changes from **90.08 %**, **89.63 %**, **90.79 %** to **91.55 %**, **90.36 %**, **91.41 %** respectively.

4.4. Third Substudy: Impact of Learning Rate and Weight Decay on Imbalances in Branch Reliance

The following study explores the impact of selected hyperparameters values on the extent to which the S-ResNet-18 model relies on the weights of its individual branches during forward propagation, measured as the ratio $\frac{\text{Tr}(\mathbf{F}_l)}{\text{Tr}(\mathbf{F}_r)}$, when trained on the SBR-CIFAR10 dataset, where one half of each image is blurred, making it less informative than the other half.



Figure 4.7: Classification test accuracy of the S-ResNet-18 model on the SBR-CIFAR10 dataset depending on the values of learning rate and weight decay.

Figure 4.8 presents the experiments arranged in two columns. In the first column, the plots have a fixed weight decay value, while the curves for different learning rate values are compared. In the second column, the opposite situation occurs with varying weight decay and a fixed learning rate.

As evident from the plots, in each case, an increase in either the learning rate or weight decay results in a more pronounced imbalance in the model’s reliance on the branch weights for the higher-quality half of the image, compared to the other. Comparing these findings with the results from Figure 4.7, it becomes apparent that a smaller imbalance is not correlated with improved final test classification accuracy.

This is particularly intriguing because, as demonstrated in Jastrzebski et al. 2021, a higher learning rate is correlated with a smaller trace of the Fisher Information Matrix over the course of training. Interestingly, it also appears to correlate with smaller traces of the the Fisher Information Matrix for each branch, a result that extends beyond the scope of this

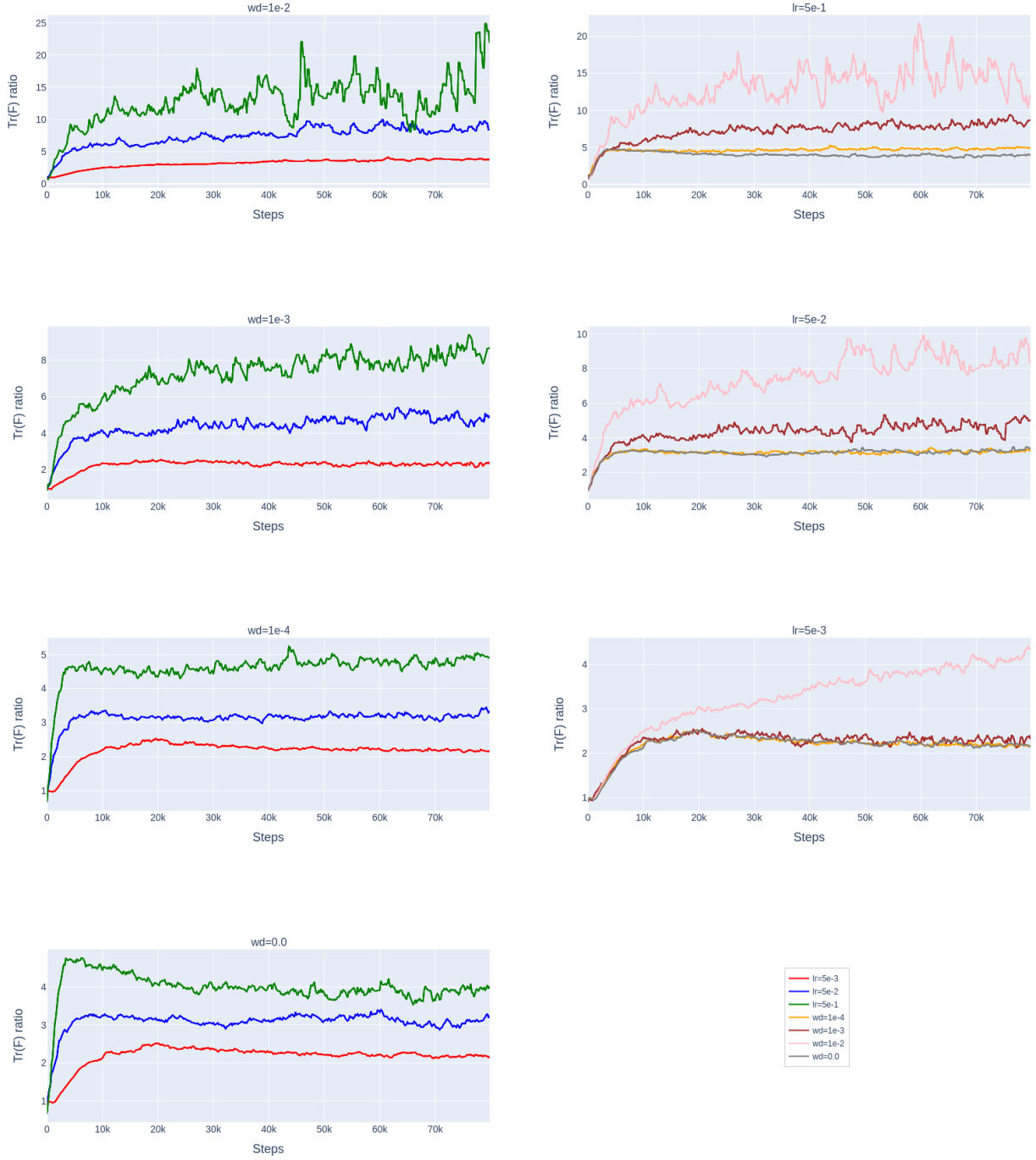


Figure 4.8: Comparison of the dynamics of the ratio of $\text{Tr}(\mathbf{F})$ s depending on the set hyperparameter. The left column groups different dynamics for learning rate, while the right column groups different dynamics for various values of weight decay. In nearly every case, we observe that an increase in the value of one hyperparameter, while keeping the other constant, leads to a higher value of $\frac{\text{Tr}(\mathbf{F}_1)}{\text{Tr}(\mathbf{F}_r)}$, except in situations where the graphs overlap. Additionally, in most cases, the value of the ratio is established early in the training stage.

work; hence, graphics illustrating this phenomenon have been omitted from this paper. Hence, let it be considered as a hypothesis. If it is true, then the phenomenon of the increasing ratio of these two values during training, with an increase in learning rate, results from either a faster decrease in $\text{Tr}(\mathbf{F}_r)$ compared to $\text{Tr}(\mathbf{F}_l)$, a slower decrease in $\text{Tr}(\mathbf{F}_l)$ compared to $\text{Tr}(\mathbf{F}_r)$, or both decreasing at the same rate, depending on how quickly this ratio increases, with the increase of learning rate, which is a possible direction for further research on the dependence of dynamics on hyperparameters.

Observing the preservation of branch imbalance from phase one through phase two, as discussed in Section 4.2, it can be hypothesized that this dynamic will maintain its pattern during further training on the S-CIFAR10 dataset. We can conclude that a smaller imbalance in branch reliance is not strongly correlated with improved final classification accuracy, as observed in the correlational study in Section 4.2. Therefore, the notion that better final classification accuracy is achieved when the ratio between the two branches $\frac{\text{Tr}(\mathbf{F}_l)}{\text{Tr}(\mathbf{F}_r)}$ is close to 1 remains applicable only within the confines of the established hyperparameters.

A similar observation regarding the impact of regularization on the magnitude of the imbalance in branch reliance was noted in the study by Wu et al. 2022, where researchers demonstrated that a higher coefficient for L1 regularization encourages the model to more greedily rely on the modality that provides more relevant information with respect to the classification task.

4.5. Additional Substudy to Main Study: Another Models and Datasets

The final study of this chapter complements the main study by presenting it on a different model and with a different dataset, as well as applying modifications to the training procedure used in the main study.

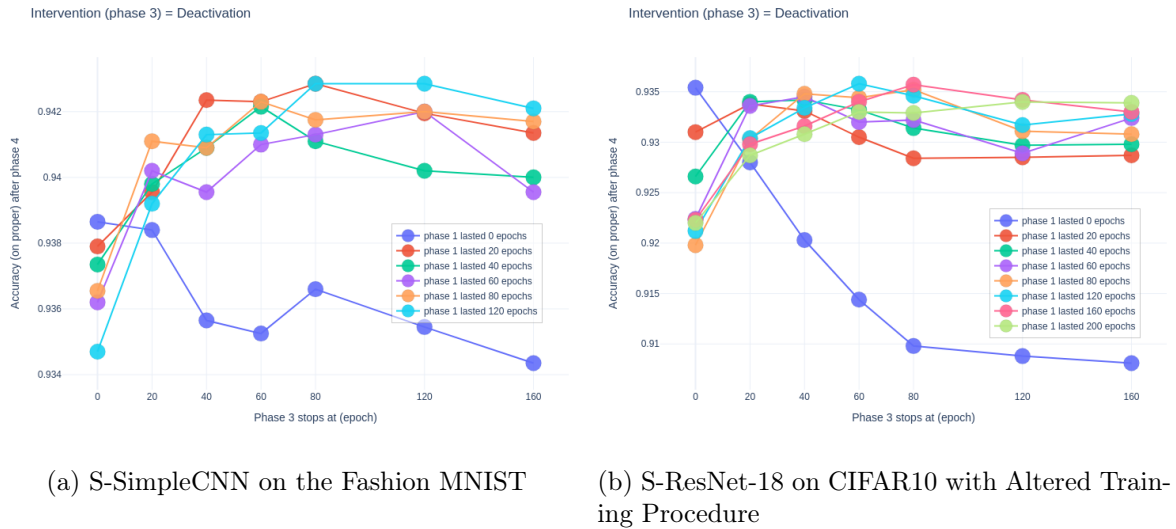


Figure 4.9: Repeating the main study, when the intervention is conducted immediately after the deprivation phase, in a different setting confirms that applying the appropriate intervention leads to a correction in test classification accuracy at the end of fourth phase.

Figure 4.9 shows results for interventions applied immediately post-deprivation phase. Like the main study, each training break in phase one has a corrective intervention length for phase 4 test accuracy. Test accuracy declines with extended phase 1 (when phase 3 lasted **0** epochs) and longer interventions (when phase 1 lasted **0** epochs), except for individual cases that disrupt the monotonicity.

Chapter 5

Discussion

5.1. Summary of Main Experimental Findings

This discussion begins by summarizing the principal findings from the experiments:

- The proposed intervention of appropriate length results in the correction of accuracy.
- The intervention of such appropriate length leads to proper activation of both branches of the bimodal network.
- A correlation is presented between the final classification effectiveness on the test dataset and how the model activates each branch during forward propagation in training.
- An appropriate length of intervention restores an even distribution of neurons in terms of sensitivity to inputs for each branch.
- The imbalance between branches can be modulated by hyperparameter changes alone, which does not translate into a correlation with accuracy as observed in the study within established hyperparameters; therefore, balancing alone cannot be a definitive indicator of whether training will be successful.
- Preliminary studies were conducted on another dataset with a different model, and the main study was expanded to infer that the main findings are not limited to a specific model and dataset.

5.2. Conclusions

In the literature, studies like Wu et al. 2022 have referred to the phenomenon where an information imbalance between modalities in multimodal DNNs leads to the predominant use of information from one modality over others. However, to the best of the author’s knowledge, none have conducted a systematic investigation of this phenomenon, even in such a simple setting as presented here, where two modalities have a similar input distribution, demonstrating that the model’s performance on a specific dataset, in this setting, is related to the balancing of the quality of the sources of the considered modalities.

Although the study mentioned in Wu et al. 2022 suggests a preventive algorithm that ensures DNNs learn from different modalities at the same pace, it does not specify whether the algorithm is effective if not implemented from the start of the training. The efficacy of applying

this algorithm at various stages of training, especially with imbalances between modalities, presents an interesting avenue for further research, even in the context of the problem considered in this work.

The current study emphasizes a method that corrects network behavior both at an early training stage and when the network converges on corrupted data. However, a definite drawback of the proposed method in this work, which involves deactivating one of the branches, is the lack of specification regarding the duration of this deactivation to correct the model's performance. This is especially important considering that an intervention lasting an epoch more or less than the best-found duration yielded noticeably poorer final classification results. Additionally, if necessary, we would also like to conduct interventions with precision to training steps, rather than whole epochs.

For this very reason, to utilize all available modalities or to mitigate the risks associated with data corruption or other anomalies, not only in the initial learning phase but also beyond it, there exists an urgent need for a comprehensive algorithm, either preventive or interventional. This algorithm could employ real-time monitoring and adaptive adjustments based on pre-defined performance metrics. It is also critically needed for correcting the model's ability to restore neural network plasticity during the learning of subsequent tasks [Zhuang et al. 2020, Lange et al. 2021].

Previous studies have indicated the existence of so-called "critical periods" in neural network learning [Achille et al. 2019, Kleinman et al. 2022]. The current research confirms these periods. However, it should be noted that these periods are not as decisive for the final classification efficiency in bimodal DNNs as previously assumed [Kleinman et al. 2022]. The current study suggests a method for correcting the network to a state that allows for both classification and distribution of unit activity in the layer, as measured using RSV, similar to that in networks not affected by the deficit. Nevertheless, the dynamics of the formation of critical learning periods in multi-modal or uni-modal scenarios still require further explanation.

Although the connections drawn between biological and artificial information processing systems are quite loose and require detachment from the objects implementing them on both sides of the theory, they seem to serving as additional ways to describe complex processes occurring in both fields of science, and above all, driving these fields forward.

5.3. Limitations and potential further study

It should be noted that in this work, the only hyperparameter subjected to tuning, besides Substudy 4.4 and Substudy 4.5, is the learning rate; all other hyperparameters are fixed to avoid introducing additional factors into the training. This approach is chosen because, as the author believes, preliminary studies should be conducted with the simplest training procedures to facilitate comparison with results obtained using additional regularization methods. Further such investigation with hyperparameter tuning, such as weight decay, momentum, the use of a scheduler, or employing other optimization methods should be considered. It should also be considered that in this study, the relationship between the data fed into each branch is highly specific.

It is necessary to investigate how other methods of data fusion from two branches, such as concatenation rather than simple summation, affect the observed phenomena. Other ways

of information flow in a bimodal network, such as intermediate fusion with a Multimodal Transfer Module, which is presented in Joze et al. 2020, should also be explored.

Another intriguing question is whether the application of the preventive algorithm from the study Wu et al. 2022 in the context of the current study results in a correction of classification effectiveness when applied immediately after the first or second phase of training.

The most critical research question, however, is what happens during the intervention being conducted. One of the potential initial steps to investigate this area could be methods visualizing the weight trajectories of both branches in the parameter space for branches with the same architecture, throughout the entire training. Focusing specifically on this stage seems like an appropriate approach to determine whether the branch weights are directed towards those of a model not trained on a dataset with corrupted data, or whether they find completely new weights in this way.

As previously mentioned in Section 5.2, a drawback of the demonstrated intervention is the difficulty in deciding in advance how long it should last, hence the need for its automation if there is any necessity to employ this method.

Finally, it remains to mention the need to check the discussed phenomenon on a larger number of models with different datasets, as well as other optimization methods.

5.4. Source code

The code for these experiments has been released to the public and is accessible via GitHub at <https://github.com/BartekKrzepkowski/CLPIntervention>.

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